DEVELOPMENT OF WEB-BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT

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Gerod spends his free time playing video games, an activity that helps him relax while also sharpening his focus, strategic thinking, and teamwork. Even when he faces failure or challenges, he keeps moving forward and learns from every experience. This mindset helps him stay strong, keep going, and never give up on his goals.

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Mario spends his free time playing video games, an activity that helps him relax while also sharpening his focus, strategic thinking, and teamwork. Even when he faces failure or challenges, he keeps moving forward and learns from every experience. This mindset helps him stay strong, keep going, and never give up on his goals.

It always seems impossible until it's done. This quote holds meaning for him because it reflects the perseverance he brings to his studies and goals, where challenges that first seem too big become achievable through patience, effort, and consistency.

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ABSTRACT

ARUTA, GEROD S. BERTILLO, CHRISHAWNE OMER E. BULLECER, MARIO F. and VILLAGRACIA, JAKE R. Development of Web-Based Community Management System for HappYness Project. Undergraduate Capstone Project. Bachelor of Science in Information Technology. Cavite State University – Bacoor Campus, Bacoor, Cavite. October 2026. Adviser: Ms. Donnalyn B. Montallana, MIT.

The study aimed to develop a Web-Based Community Management System for the HappYness Project. It was designed to help the organization centralize and manage its operations across its project sites in Bacoor, Quiapo, and Binangonan. The application brings together volunteer tracking and performance monitoring, activity scheduling and monitoring, donation inventory control, beneficiary administration, document automation for reports and certificates, an e-commerce platform for community products, and a real-time reporting dashboard. It also provides an administrator interface for managing core community tasks, user accounts, and system data.

The researchers developed the web-based application using HTML5, CSS3, and JavaScript (ES6) for the front-end interface, Python 3.10 with Django for back-end processing, and SQLite3 for database management. System requirements and user feedback were collected through interviews, observation, and fishbone analysis of existing manual workflows. Unit, integration, and system testing were conducted to evaluate functional suitability, performance efficiency, usability, and data security.

The application was evaluated using an instrument based on the ISO 25010 software quality standards by 48 respondents, consisting of 30 beneficiaries, 8 admin/staff members, and 10 IT professionals. The system achieved an overall mean rating of 4.09, interpreted as "Very Good," indicating that the platform successfully automates manual administrative processes, improves operational transparency, and meets the quality requirements for organizational deployment.

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DEVELOPMENT OF WEB- BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT

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INTRODUCTION

Community-based and volunteer-driven initiatives are essential for fostering local development and supporting marginalized populations worldwide. However, many of these organizations struggle with traditional, manual workflows that lead to administrative bottlenecks, data discrepancies, and communication delays, ultimately limiting their social impact.

To overcome these inefficiencies, centralized web-based information systems offer a digital solution to automate administrative tasks, optimize volunteer tracking, and ensure real-time transparency. A prime beneficiary of such a system is the HappYness Project Bacoor, an SEC- and BIR-registered charitable organization established in December 2018 in Bacoor, Cavite, Philippines. Providing essential goods like groceries, school supplies, and clothing, the organization has executed nearly 1,000 outreach programs for public schools, hospitals, orphanages, and over 80 indigent communities nationwide.

Despite its extensive reach, coordinating a growing volunteer network and maintaining manual records present ongoing operational hurdles. To address these challenges, this study presents a Web-Based Community Management System tailored for the HappYness Project. By centralizing core activities—such as donation management, volunteer tracking, teaching, planting, and rag-making—into a secure digital platform with real-time reporting dashboards, the proposed system aims to enhance operational efficiency, ensure data integrity, and strengthen collaboration among stakeholders.

Project Context

The HappYness Project is an SEC- and BIR-registered charitable organization established in December 2018 in Bacoor, Cavite, Philippines, dedicated to uplifting underprivileged children and families through essential goods, education, and social welfare programs. The organization handles a steady volume of community transactions across its primary sites in Bacoor, Quiapo, and Binangonan. These daily transactions include managing incoming and outgoing donations (such as groceries, shoes, slippers, and school supplies), tracking volunteer participation, and executing diverse community outreach activities such as teaching children, planting, and rag-making.

However, as the organization continues to expand, its reliance on manual methods for processing these transactions creates operational challenges, including inefficiencies, delays, data inconsistencies, and difficulty maintaining accurate records. To resolve these bottlenecks, a Web-Based Community Management System has been developed to streamline and centralize operations.

The proposed system integrates key modules for volunteer management, activity tracking, and donation coordination, supported by a real-time reporting feature. This functionality eliminates manual data entry by sending automated queries to the SQLite3 database to retrieve the latest project metrics, which are then visualized through real-time dashboards providing live data on resource levels, volunteer participation, and program status. Furthermore, the integration of document automation allows for the instant generation of PDF reports and digital participation certificates, enhancing accountability for staff and student volunteers engaged in on-the-job training (OJT). Ultimately, the system is designed to promote collaboration and sustainability, enabling the HappYness Project to scale its operations effectively.

Objectives of the Study

To develop a Web-Based Community Management System for the HappYness Project that enhances operational efficiency and strengthens collaboration among volunteers, staff, and community members.

Specifically, the study aims to:

1. Design and develop a centralized web-based system that automates the core administrative workflows of the organization, specifically through:
    a. Activity Management: Automating the scheduling, status tracking, and logging of community tasks such as rag-making, planting, and teaching.
    b. Volunteer Coordination: Forms for registration, digital attendance, and automated task assignment for students and community members.

    c. Donation Inventory Control: A digital ledger tracking incoming donations and their distribution to indigent communities

2. Develop the system using a standardized tech stack to ensure accessibility and data persistence:
    a. Front-end: HTML5, CSS3, and JavaScript (ES6) for a responsive interface.
    b. Back-end: PHP 8.0+ or Python 3.10+ to handle the system's logic and automation.
    c. Database: SQLite3 for structured and reliable storage of project records.
    d. Deployment**:** A secure web hosting platform to allow remote access for staff in Bacoor, Quiapo, and Binangonan.
3. Incorporate automated monitoring and reporting features to enhance organizational transparency:
    a. Real-Time Dashboards: To visualize project impact metrics and resource levels without manual data entry.
    b. Document Automation: To generate instant PDF reports and digital participation certificates for volunteers.
4. Conduct rigorous system testing to evaluate performance, reliability, and process automation:
    a. Unit and Integration Testing**:** Ensuring individual modules and components integrate seamlessly.
    b. System Testing**:** Verifying that automated workflows successfully replace manual processes.
5. Evaluate the system against ISO 25010 Software Quality Standards focusing on:
    a. Functional Suitability: Assessing how effectively the platform automates intended tasks.

   b. Usability: Measuring how easily staff and volunteers with varying technical skills can navigate the platform.

6. Formulate a Sustainable Implementation Plan that includes:
   a. User Training Programs**:** Ensuring smooth adoption by project staff.
   b. Maintenance Guidelines: Supporting long-term system and database management.

Purpose and Description

The purpose of the study is to design and implement a Web-Based Community Management System for the HappYness Project. This study is developed in response to the operational limitations experienced by the organization, particularly in managing volunteers, coordinating donations, and monitoring multiple community-based activities. The current reliance on manual processes often leads to inefficiencies, delayed communication, data inconsistencies, and difficulty in maintaining centralized and accurate records, which negatively affect the overall efficiency and scalability of the project.

To address these challenges, the system aims to streamline operations, improve coordination, and enhance collaboration among volunteers, staff, and beneficiaries through a centralized digital platform. By integrating core activities such as rag-making, planting, teaching children, and donation outreach into a single system, the platform is designed to improve organization, improved accessibility of information, and real-time monitoring of activities.

Furthermore, the system automates administrative processes to reduce workload and minimize human error while ensuring transparency and accountability through structured reporting mechanisms. It also provides tools

for role allocation, task monitoring, and collaboration, which help improve engagement and productivity among social work students and community volunteers.

In addition, the system integrates modules such as interactive tutorials, event scheduling, inventory tracking, and activity monitoring to support efficient and organized project execution. These features are designed to address gaps in coordination and improve overall workflow management within the organization.

Lastly, the system is designed with scalability in mind, allowing future enhancements such as e-commerce integration for community-made products, automated certificate generation, and donor reporting tools. Through these capabilities, the system supports the long-term sustainability of the HappYness Project by leveraging technology to enhance efficiency, transparency, and community impact.

Time and Place of the Study

The development and implementation of the Web-Based Community Management System for the HappYness Project will begin in February 2026 with the initial planning and requirement-gathering phase in Bacoor City, Cavite. Following this, the design and development stages will be conducted throughout March 2026, also centered in Bacoor City, to establish the system's architecture and core features. From April to June 2026, the system will undergo rigorous testing in the same location to ensure functionality and reliability.

By July 2026, the study will expand its reach to include Quiapo and Binangonan alongside Bacoor City for the feedback collection phase. This will

allow the researchers to gather insights from a broader range of users across the primary areas where the HappYness Project operates. The final deployment and implementation of the system will take place in August 2026 across all three locations, supporting core activities such as rag-making, planting, and educational programs. By piloting the system in these diverse communities, the study will ensure that the platform effectively addresses the operational needs of social work students and beneficiaries while fostering meaningful collaboration across all project sites.

Scope and Limitations

This study focuses on the design, development, and implementation of a Web-Based Community Management System for the HappYness Project to improve operational efficiency, enhance communication, and support core activities. The system includes key features such as volunteer management, activity tracking, donation management, and automated reporting. Additional functionalities such as certificate generation for volunteers, an e-commerce module for selling community-made products, and a donation portal are integrated to provide sustainability and transparency. The study specifically caters to the needs of the HappYness Project's teams and community members in Bacoor City, Quiapo, and Binangonan, ensuring that the platform aligns with the organization's existing outreach programs, which restricts its immediate applicability to other communities with different operational structures and needs.

Despite its comprehensive scope, the study has several limitations. The development period restricts the inclusion of advanced features such as mobile application compatibility. Budget constraints limit access to premium cloud hosting, advanced security protocols, and third-party API integrations

that could enhance system performance. Additionally, internet connectivity challenges and varying levels of technical proficiency among volunteers and beneficiaries may affect system adoption, requiring additional training and user support.

The system's evaluation is limited to user testing, feedback collection, and the application of ISO 25010 software quality standards. As such, long-term performance outcomes such as improvements in volunteer retention, donation growth, and overall operational efficiency cannot be fully measured within the timeframe of this study and may require further research for comprehensive assessment. Nevertheless, the system is designed with scalability and adaptability in mind, allowing for future enhancements and expansion in accordance with the evolving needs of the HappYness Project.

Conceptual Framework

DEVELOPMENT OF WEB- BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT

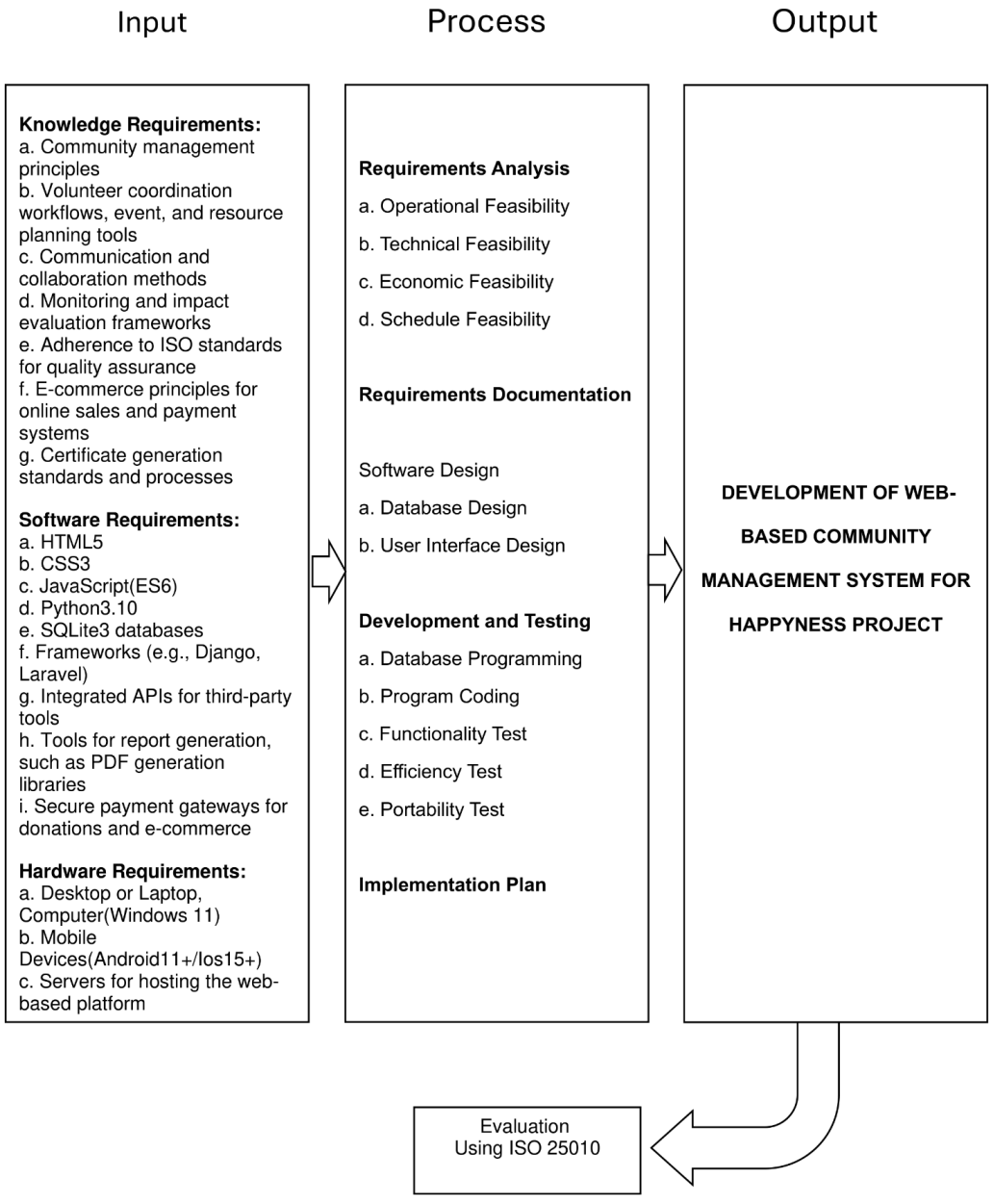


Figure 1. Conceptual Framework of DEVELOPMENT WEB- BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT

The input includes the existing operational challenges of the HappYness Project such as manual processes in volunteer management,

donation tracking, activity monitoring, and communication gaps. It also includes user requirements from volunteers, staff, and beneficiaries, as well as system requirements necessary for the development of a web-based platform. The process outlines the steps taken to develop the system, including requirement gathering, system analysis, documentation, system design, database design, system development, testing, and deployment. These steps lead to the output, which is a Web-Based Community Management System for the HappYness Project designed to centralize operations, improve coordination, and support core activities such as rag-making, planting, teaching children, and donation outreach.

Definition of terms

The following terms are operationally defined in the study:

Activity Management Module – This is the specific part of the system that handles project tasks like rag-making and teaching. It allows users to perform CRUD operations—meaning they can Create new tasks, Read or view them, Update progress, and Delete completed items in the database.

Community Management System (CMS) – The main web-based tool created to organize the project's data in one place. It acts as a Management Information System (MIS) that uses a Client-Server architecture, where the website (the client) talks to a central computer (the server) to automate work that used to be done by hand.

Database Management (SQLite3) – The Back-End storage area where all project information is saved. It provides Data Persistence (making sure data isn't lost when the power goes off) and Data Integrity (keeping information organized and correct) using structured tables.

Django Framework – A high-level Python web framework that follows the Model-Template-View (MTV) architecture. It provides "batteries-included" features, such as an automatic admin interface and built-in security, to accelerate the development of complex, data-driven websites.

Donation Management Subsystem – A dedicated section within the system that keeps track of resources like money or supplies. It uses relational database tables (linked lists of data) to log what was donated and who received it, making sure the inventory is accurate and transparent.

Python 3.16 – A high-level, versatile programming language known for its readability and efficiency. It serves as the primary language for the system's logic, processing, and integration with the Django framework.

Real-Time Reporting and Data Visualization – A feature that sends Queries (requests for info) to the database to get the latest numbers. It turns raw data into Dashboards (visual charts) and PDF files to help the organization make better decisions based on actual facts.

Volunteer Tracking and Performance Monitoring – A feature that manages User Authentication (logging in) and Role-Based Access Control (RBAC), which ensures users only see what they are allowed to. It records a volunteer's hours and work to automatically create reports and digital certificates.

Web-Based Interface – The Front-End or the actual "face" of the website that users interact with. It was built using HTML5, CSS3, and JavaScript to be a Responsive User Interface (UI), meaning the design automatically shrinks or expands to look good on both phones and computers.

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of related literature and studies that provide the foundation for the development of the Web-Based Community Management System for the HappYness Project. It explores existing research on technology-driven community management, the role of web-based platforms in nonprofit initiatives, and key technical considerations for system development.

Local Related Literature

Digitalizing Governance: A Transformation on the Processes of One Community in the Philippines. This study examined how digital governance tools reshape the management processes within local communities in the Philippines. It emphasized that integrating web-based systems enhances accountability, transparency, and citizen participation in local projects. Through interviews and system analysis, the researchers found that shifting from manual operations to digital platforms significantly improved data accessibility and operational efficiency. The study also revealed that adopting such systems reduces human error and promotes faster communication between departments and community members. However, it highlighted challenges such as inadequate training and limited access to technology in rural areas, which hinder full adoption. The findings suggest that a web-based management system can be a vital tool for organizing community activities, tracking initiatives, and maintaining transparency among stakeholders in local organizations. (Velarde, 2022)

Advancing Disaster Resilience in the Philippines: Blueprint for Education Extension Initiative. This study explored the role of education and technology in enhancing disaster resilience within Philippine communities. The

researchers proposed the integration of web-based management tools to strengthen coordination among volunteers, educators, and local authorities. The study showed that these systems facilitate faster dissemination of information and improve preparedness through data-driven strategies. Furthermore, the research revealed that online coordination platforms enhance accountability and ensure better utilization of community resources. By providing a structured method for documentation and collaboration, digital tools help establish a more organized and responsive system for community-based disaster management. The findings underscored that such technological initiatives empower both organizations and citizens to act efficiently during emergencies. (Torino et al., 2025)

Community Needs Assessment as Basis for the Extension Program of Philippine College Foundation. This research emphasized the importance of conducting community needs assessments to design effective extension and outreach programs. It revealed that many local institutions rely on manual data gathering, which leads to inefficiencies and inaccuracies in identifying community issues. The study proposed adopting web-based management systems to streamline data collection, storage, and analysis. By integrating such platforms, organizations can easily monitor community needs, track program progress, and generate reports for evaluation. The findings also showed that a centralized database promotes coordination among staff and volunteers, ensuring more organized and targeted interventions. Ultimately, this research demonstrated that digital tools enhance the responsiveness and sustainability of community-based programs. (Dela Cruz, 2023)

Compliance to Republic Act 11292: The Seal of Good Local Governance Act of 2019 in Selected Municipalities in One Philippine Province. This study analyzed how municipalities comply with the Seal of

Good Local Governance Act and how digital systems support transparency in implementing public programs. The researchers observed that municipalities using web-based documentation systems exhibited better compliance and accountability. These platforms improved monitoring of local government projects, financial reporting, and citizen engagement. However, the study also highlighted the need for digital literacy training to ensure that local officers can effectively use such tools. Findings suggested that integrating information systems in community operations strengthens governance performance and public trust. This research provided a solid foundation for understanding how online management systems can enhance community transparency and service delivery. (Mendoza et al., 2023)

Strengthening Disaster Response: A Scoping Review of Communication and Coordination Training Programs in the Philippine Higher Education Institutions. This scoping review focused on how communication and coordination strategies can be enhanced through digital tools in Philippine educational and community settings. It found that web-based systems play a crucial role in improving coordination, data management, and rapid response during crises. The study identified the need for centralized platforms that store essential information and streamline communication between departments and partner organizations. Results demonstrated that adopting web-based management systems reduces delays and ensures a more efficient information flow. It concluded that digital transformation in coordination processes is vital to strengthening both disaster preparedness and community engagement. (Castillo et al., 2024)

Foreign Related Literature

Technology Integration and Digital Transformation in Non-Profit Organizations. This study explored how technology integration transforms

operations in nonprofit organizations globally. It examined various digital tools that optimize communication, automate administrative processes, and improve donor engagement. The researchers found that nonprofits adopting digital systems experienced significant improvements in operational efficiency, volunteer participation, and transparency. The paper also emphasized the importance of leadership support and continuous staff training for the success of technological adoption. Moreover, it discussed the challenges of digital inequality and resistance to change among employees. The findings concluded that a well-implemented digital management system contributes to sustainable growth, trust, and innovation within nonprofit institutions worldwide. (Hughes et al., 2024)

The Future Role of Non-Profits: Digitalization, Grassroots Renewal, and Emerging Paradigms of Civil Society. This research analyzed how digitalization reshapes the landscape of nonprofit organizations and civic engagement. It discussed how web-based platforms enhance collaboration, transparency, and public participation across social causes. The authors presented case studies demonstrating how NGOs used digital systems to improve project coordination, financial accountability, and volunteer mobilization. The study also noted that digitalization empowers grassroots movements by giving communities a platform to voice their needs and track organizational impact. It concluded that technology-driven transparency fosters stronger relationships between nonprofits and their stakeholders, encouraging long-term trust and sustainability. (Harrison, 2024)

NPO Web-Based Accountability: How Can We Know If NPOs Are Doing Good Things? This study examined the concept of accountability in nonprofit organizations through the use of web-based transparency systems. It explained how online platforms allow organizations to disclose financial

reports, project results, and performance metrics to the public. The findings indicated that digital accountability systems help build donor trust and promote ethical governance. Additionally, the study noted that web-based reporting improves communication with stakeholders and enhances organizational credibility. It emphasized that adopting digital systems is essential for maintaining transparency and for demonstrating the social impact of nonprofit activities. The research concluded that online accountability mechanisms are now considered a best practice in modern nonprofit management. (Wang, 2021)

Digital Transformation: A Meta-Review and Guidelines for Future Research. This meta-review analyzed existing studies on digital transformation, offering guidelines for its successful implementation across sectors, including nonprofits. The research highlighted how digital transformation enables better data management, coordination, and decision-making in organizations. The authors emphasized that organizations that embrace web-based systems gain a competitive advantage through real-time analytics and improved efficiency. They also outlined the key factors influencing digital success, such as leadership support, resource availability, and technological adaptability. This comprehensive review provides insights into how digital innovation drives organizational sustainability and long-term development, making it a valuable reference for system designers and social organizations alike. (Chen & Park, 2023)

Design and Implementation Factors for Performance Measurement in Non-Profit Organizations: A Literature Review. This study reviewed critical factors that influence performance measurement systems in nonprofit organizations. It emphasized that digital tools enable more accurate tracking of outcomes, resource utilization, and program effectiveness. The

authors discussed how web-based platforms improve transparency and accountability by automating data collection and performance reporting. They also suggested that aligning these tools with organizational goals enhances decision-making and impact assessment. The findings highlighted that integrating technology not only simplifies management processes but also fosters innovation and data-driven culture in nonprofits. The study concluded that web-based systems are essential for achieving measurable and sustainable results in modern organizations. (Almeida, 2023)

Review of Related Studies

Local Related Studies

Website Development for Religious Non-Profits: Accessibility, Optimization, and Community Engagement. This study focused on developing a responsive and accessible website for religious nonprofit organizations in the Philippines to improve community communication and participation. The system was designed to help organizations disseminate announcements, manage donations, and promote events efficiently. Using user experience (UX) design principles and SEO optimization strategies, the researchers tested the system's accessibility in both urban and rural settings. The results showed significant improvements in member engagement, donation transparency, and access to organizational updates. Additionally, the study highlighted how responsive design allows users to interact with the system through multiple devices, enhancing inclusivity. The project concluded that web-based platforms play a vital role in strengthening the relationship between organizations and their communities, especially for nonprofits that rely on volunteer-driven participation. (Reyes et al., 2025)

Mobile Technology for Improving Project Management in the Construction Industry – Philippine Setting. This study explored how mobile-based platforms improve project management and team coordination in the Philippine construction industry, offering valuable insights into community-based project administration. The system allowed teams to track project progress, monitor deadlines, and share updates in real time. Researchers found that adopting mobile and web-based management systems resulted in improved communication, reduced project delays, and increased worker accountability. The study also identified that user-friendly dashboards and mobile compatibility encouraged participation even among users with limited digital literacy. Findings supported that mobile technology bridges gaps in coordination, ensuring more efficient and transparent management of community or organizational projects. This aligns with modern efforts to digitize management systems for better collaboration and operational success. (Alonzo et al., 2024)

Digital Transformation and Management Systems in Non-Profit Organizations. This study investigated the role of integrated web-based systems in improving the operational capacity of non-profit sectors. By evaluating platforms designed for resource allocation and stakeholder management, the research found that digitizing administrative workflows—such as beneficiary tracking and donation logging—significantly minimizes data redundancy and human error. The findings highlight that real-time data accessibility fosters greater institutional transparency, which is critical for maintaining donor trust and community credibility. Furthermore, the study concluded that user-centric design in these systems is a primary driver for volunteer retention and effective service delivery. This research underscores that for non-profits to scale their social impact, they must

transition from manual processes to automated, data-driven frameworks. (Alonzo & Reyes, 2024)

Digital Barangay: Ayos Lomboy's Web-Based Management and Information System. This research addressed the challenges of traditional, manual record-keeping in local Philippine communities by developing a centralized web-based management system. The study focused on automating resident profiling and administrative workflows to eliminate data redundancy and improve the speed of service delivery. Castillo and his colleagues found that the transition to a digital platform allowed for more accurate data handling and provided community leaders with real-time insights into resident needs. This study is highly relevant to your project as it highlights how web-based systems can replace fragmented manual processes in a local, resource-constrained setting. (Castillo et al., 2025).

Volunteerism in the Philippines: A Systematic Literature Review. This comprehensive review analyzed the landscape of Filipino volunteerism over the last decade to identify the factors that drive long-term engagement. Cruz and Mendoza identified a critical gap between volunteer intent and organizational capacity, noting that many local organizations lack the digital infrastructure to properly track and coordinate their members. The authors argued that the "professionalization" of volunteerism in the Philippines—moving from ad-hoc labor to structured, skill-based contributions—requires the implementation of dedicated management frameworks. Their findings support your project's objective of creating a structured Volunteer & Personnel Management module to improve organizational sustainability. (Cruz & Mendoza, 2025).

Foreign Related Studies

Empowering Change: An Advanced NGO Management System for Efficient Operations, Enhanced Donor Engagement, and Transparent Community Development. This research developed an advanced management system for NGOs to streamline project implementation, donor communication, and reporting. The system incorporated automated financial management, volunteer tracking, and real-time data visualization. Findings indicated that automation reduced administrative errors, improved task scheduling, and increased transparency in community programs. The study also revealed that the system's analytics module enhanced monitoring of organizational performance and impact evaluation. Researchers concluded that advanced web-based platforms empower nonprofits to optimize operations, foster trust with stakeholders, and sustain long-term development initiatives. The integration of technology was found to be a key driver of accountability and efficiency in modern community organizations. (Singh et al., 2023)

SmartNGO: An Integrated Platform for Managing Volunteers and Events. This study introduced SmartNGO, a web-based solution designed to automate volunteer coordination and event management in nonprofit organizations. The researchers applied an agile methodology to create a responsive and scalable system accessible across various devices. The platform included features for volunteer registration, scheduling, and feedback monitoring. Findings revealed a notable increase in volunteer satisfaction and operational efficiency, reducing manual workloads by 45%. The study highlighted the role of user-centered design in ensuring smooth navigation and participation. The system proved that well-structured digital platforms can

improve transparency and foster community engagement within NGO operations. (Joshi, D. 2024).

Development of a Web-Based Volunteer Management System for Nonprofit Organizations. This study focused on the design and development of a web-based system to manage volunteer activities, attendance, and resource allocation in nonprofit organizations. The researchers applied a structured software development process to create modules for volunteer registration, task assignment, and real-time tracking of participation. The system automated manual processes such as attendance recording and report generation, improving accuracy and efficiency. Testing results showed that the system enhanced coordination, reduced administrative workload, and improved data management. The study concluded that implementing a web-based volunteer management system significantly improves operational efficiency, transparency, and overall organizational performance (Wu et al., 2023).

Volunteer Connect: A Volunteer Coordination Platform with Chatbot Assistance via Retrieval-Augmented Generation. This study focused on developing an intelligent volunteer management system that integrates chatbot support for improved coordination and communication. The platform used Retrieval-Augmented Generation (RAG) technology to provide real-time assistance to volunteers by automating responses, task assignments, and scheduling. The study found that chatbot integration significantly reduced delays in communication and improved volunteer satisfaction. Researchers also noted that automation enhanced the efficiency of data management and reporting. The study concluded that combining artificial intelligence with web-based systems can modernize community and

volunteer operations, ensuring faster response times and better overall management. (Liang et al., 2025).

Development of a Web-Based Community Management System for Nonprofit Organizations. This study focused on the design and development of a web-based community management system aimed at improving the coordination of volunteer activities, donation management, and community programs. The researchers followed a structured software development methodology, including requirements analysis, system design, development, and testing. The system integrated key modules such as volunteer management, activity scheduling, donation tracking, and report generation to support efficient operations. The developed system automated manual processes, reducing paperwork, minimizing errors, and improving the accuracy of records. It also provided real-time monitoring and centralized data management, allowing administrators to track activities, manage resources, and generate reports efficiently. Testing results showed improved coordination among stakeholders, enhanced transparency, and better communication within the organization. The study concluded that implementing a web-based community management system significantly enhances operational efficiency, data organization, and overall project sustainability in nonprofit organizations (Garcia et al., 2024).

System Technical Background

The system is a web-based community management platform being developed for the *HappYness Project* organization. This platform is designed to efficiently manage various operations while ensuring cross-browser compatibility. The system will utilize HTML5 and CSS3 for front-end development, JavaScript (ES6) for dynamic client-side functionality, and

Python 3.10 as the backend programming language. SQLite3 will serve as the lightweight, file-based database for data storage and retrieval.

HTML5 HTML5 provides the foundational structure for the platform, enabling semantic markup (e.g., <article>, <section>) and multimedia integration without third-party plugins. Its offline capabilities via the Cache API and local storage ensure functionality even with limited connectivity (World Wide Web Consortium [W3C], 2023).

CSS3 CSS3 handles responsive design through Flexbox, Grid, and media queries, ensuring consistent styling across devices. Features like animations and transitions enhance user engagement while maintaining accessibility (Mozilla Developer Network [MDN], 2024).

JavaScript (ES6) ES6 modernizes client-side scripting with arrow functions, template literals, and let/const declarations. Its module system (import/export) organizes code for maintainability, while Promises streamline asynchronous operations like API calls (ECMA International, 2023).

Python 3.10 Python 3.10 powers the backend with features like structural pattern matching (match/case) and improved error messages. Frameworks like Flask or Django handle routing and security, while SQLite3 integration manages data storage (Python Software Foundation, 2021).

SQLite3 SQLite3 offers a serverless, zero-configuration database engine with ACID compliance. Its lightweight architecture (single-file storage) is ideal for small-to-medium workloads, such as SSG member records or election data (Hipp, 2021).

Synthesis

The reviewed literature and studies provide a strong technical and conceptual foundation for the Development of a Web-Based Community Management System for the HappYness Project. Globally, the shift toward Digital Transformation in non-profit organizations (NPOs) is no longer optional but a necessity for operational sustainability. Research by Hughes et al. (2024) and Harrison (2024) emphasizes that digitalizing manual processes directly correlates with improved volunteer participation and organizational trust. This supports the study's goal of moving the HappYness Project away from manual, paper-based tracking toward a centralized digital architecture.

A recurring theme in the literature is the critical role of Web-Based Accountability and Transparency. Wang (2021) and Almeida (2023) argue that online platforms are the most effective tools for disclosing impact metrics and financial reports to stakeholders. By integrating a Real-Time Reporting and Data Visualization feature, this project addresses the transparency gaps identified in the current manual operations of the HappYness Project, allowing for the "NPO accountability" discussed in existing research.

In the Philippine context, studies by Velarde (2022) and Mendoza et al. (2023) highlight that while digital governance significantly improves data accessibility in local communities, challenges such as technical proficiency and infrastructure remain. This study addresses these local constraints by utilizing a SQLite3 database and a responsive Web-Based Interface, ensuring the system remains lightweight, easy to deploy, and accessible even in areas with varying internet stability, such as Quiapo and Binangonan.

Furthermore, the technical feasibility of managing volunteers and resources is well-documented in the studies of Singh et al. (2023) and Patel et

al. (2024), who demonstrated that automation can reduce administrative workloads by nearly 45%. While advanced technologies like AI and RAG (Retrieval-Augmented Generation) are emerging trends in volunteer coordination (Reddy et al., 2024; Liang et al., 2025), this study prioritizes Functional Suitability and Usability through the ISO 25010 standards, focusing on a robust modular design (Activity, Donation, and Volunteer modules) that fits the current technical capacity of the organization.

Finally, the gap identified in the existing literature is the lack of a centralized, integrated platform specifically designed for grassroots community projects in the Philippines that combine educational programs, livelihood initiatives (rag-making), and donation distribution in one system. This study bridges that gap by developing a tailored Web-Based Community Management System that simplifies complex workflows into a single, secure interface, ensuring that the HappYness Project can scale its impact effectively across its multiple project sites.

METHODOLOGY

This section outlines the functional and non-functional requirements of the Web-Based Community Management System for the HappYness Project, based on community needs, best practices, and technological considerations.

Design of Software, Systems, Products and/or Processes

The HappYness Project currently relies on a fully manual process for managing community programs and beneficiaries, involving project teams, coordinators, and volunteers. The process includes distributing surveys to assess community needs, reviewing data, planning programs, registering beneficiaries, and preparing materials. After program delivery, feedback is manually collected and recorded for future improvement.

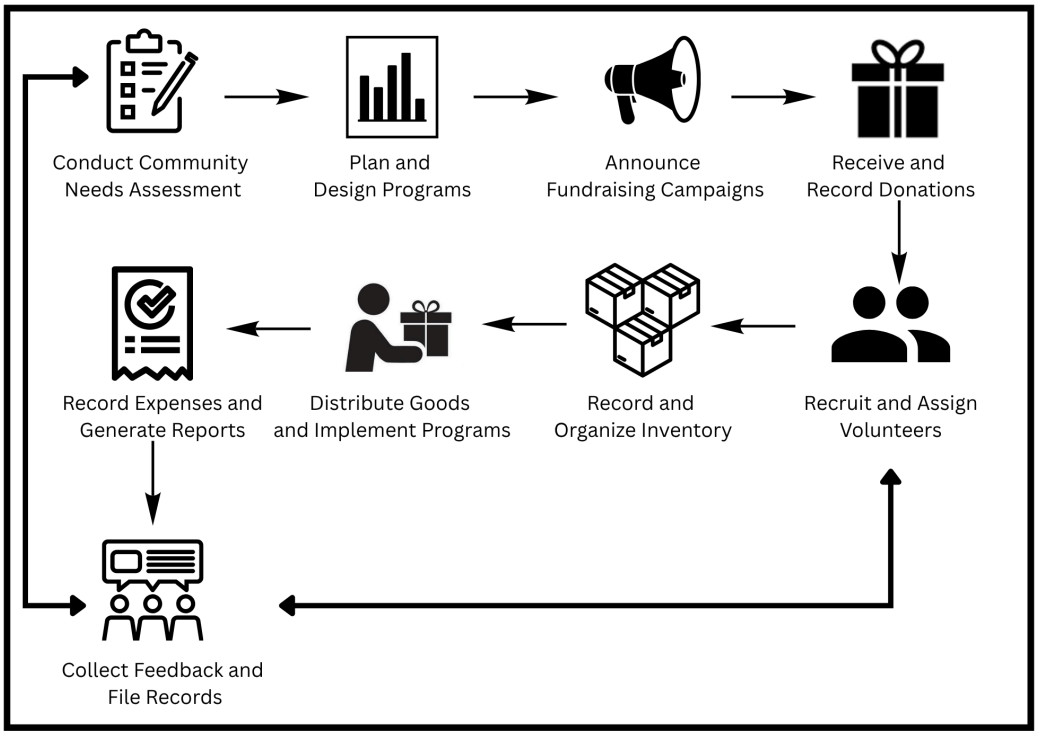


Figure 2. Manual Process Flow of the HappYness Project

Figure 2 illustrates the manual workflow of program and beneficiary management for the HappYness Project. The process begins with conducting

a community needs assessment to identify the most urgent concerns. Based on the findings, programs are planned and designed, followed by the announcement of fundraising campaigns to gather resources. Donations are then received and recorded manually, while volunteers are recruited and assigned to assist in various activities. The collected goods are recorded and organized in inventory before being distributed to beneficiaries as part of program implementation. After distribution, expenses are documented, reports are generated, and feedback is collected to evaluate program outcomes. This manual approach is highly dependent on paperwork and human coordination, which often leads to inefficiencies, data inconsistencies, and delays—emphasizing the need for a digital, automated system to streamline operations and improve transparency.

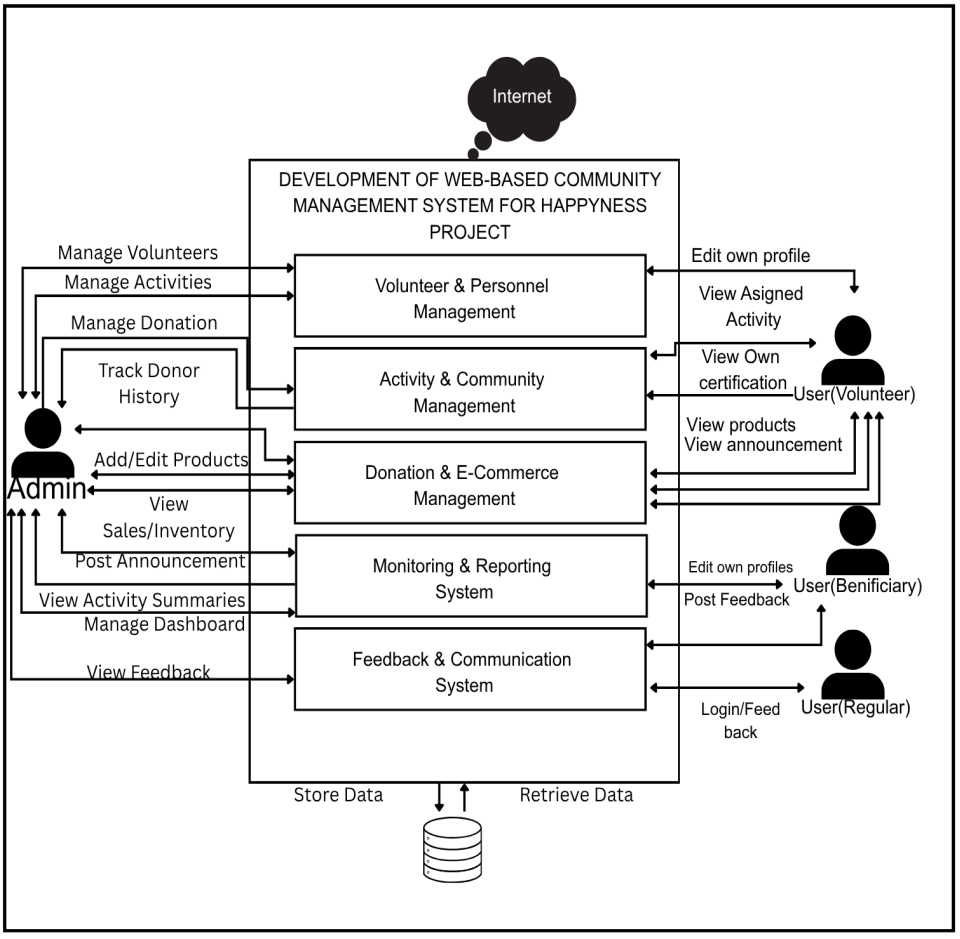


Figure 3. System Architecture

Volunteer Management. Administrators oversee volunteer registration, approval, and task assignments for various outreach activities. Volunteers can update their profiles, monitor their assigned programs, and track participation history, ensuring efficient coordination.

Activity Tracking. This module monitors the implementation of community activities such as donation drives, planting initiatives, and educational programs. Administrators can create and assign activities, volunteers can record their participation, and beneficiaries can view or register for community-based programs.

Donation Management. Handles both monetary and in-kind donations by tracking donor information, transaction details, and donation history. Administrators manage records and ensure transparency, while beneficiaries can acknowledge received aid or request assistance when eligible.

Beneficiary Management. A newly added module that allows administrators to register, verify, and manage beneficiary profiles. Beneficiaries can update their information, view distributed goods or services received, and participate in community programs.

Report Generation. Automatically compiles data from all modules—volunteer activities, donations, and beneficiary records—into summarized reports. These reports enhance transparency and decision-making for administrators and stakeholders.

E-Commerce Module. Facilitates the sale of community-produced goods such as rags, plants, and crafts. Administrators manage product listings and transactions, while beneficiaries and donors can purchase products to support the project.

Community Engagement. Provides an interactive platform for communication and feedback. Users—including beneficiaries—can send feedback, view announcements, and stay informed about upcoming initiatives and events.

Real-Time Dashboard. Displays live data on donations, volunteer participation, and program status. Administrators can easily monitor ongoing activities, track beneficiary engagement, and make data-driven decisions.

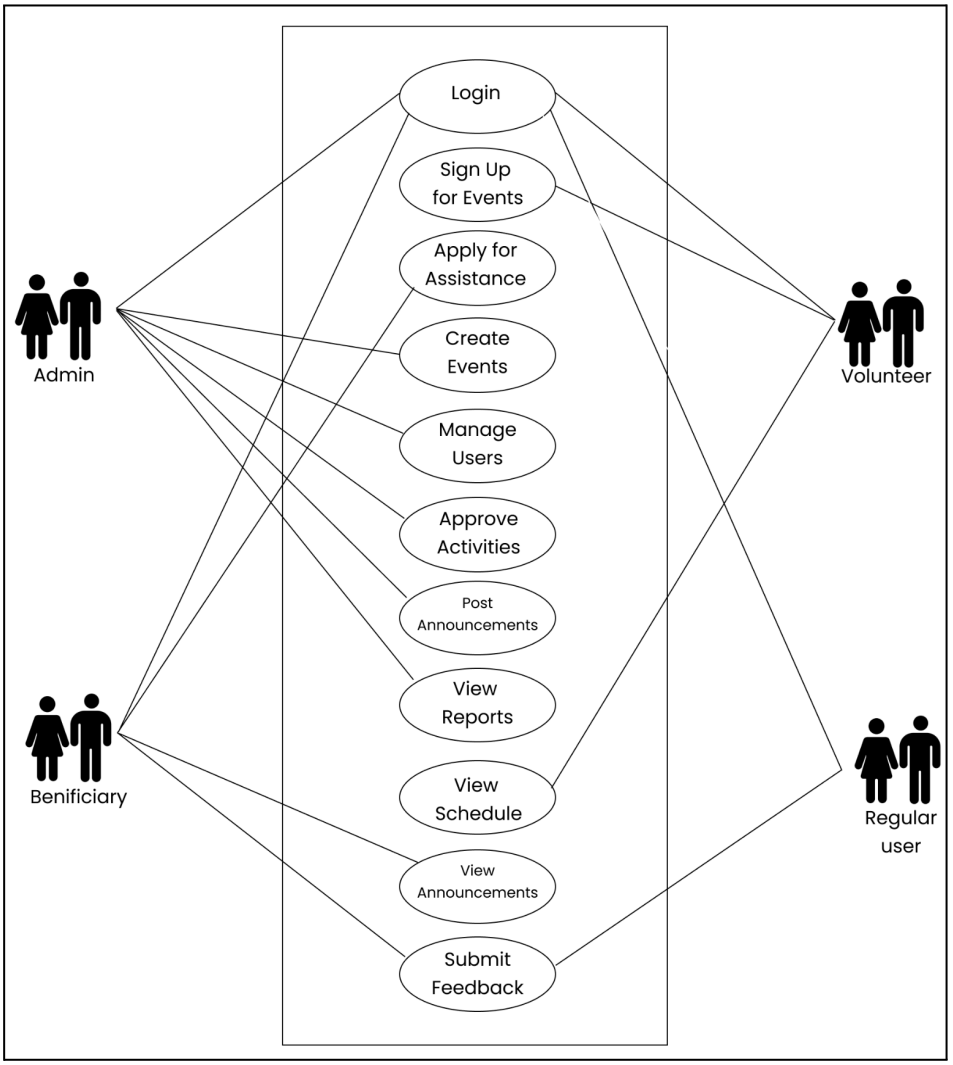


Figure 4. Use Case Diagram of the Web-Based Management System

Figure 4 shows the use case diagram for the Web-Based Community Management System, highlighting three user roles—Admin, Volunteer, and Beneficiary. All users can log in, view announcements and schedules, and submit feedback. Admins have extended privileges to create events, manage users, approve activities, post announcements, and view reports. Volunteers can sign up for events, while beneficiaries can apply for assistance. This diagram outlines how each user interacts with the system to facilitate community engagement.

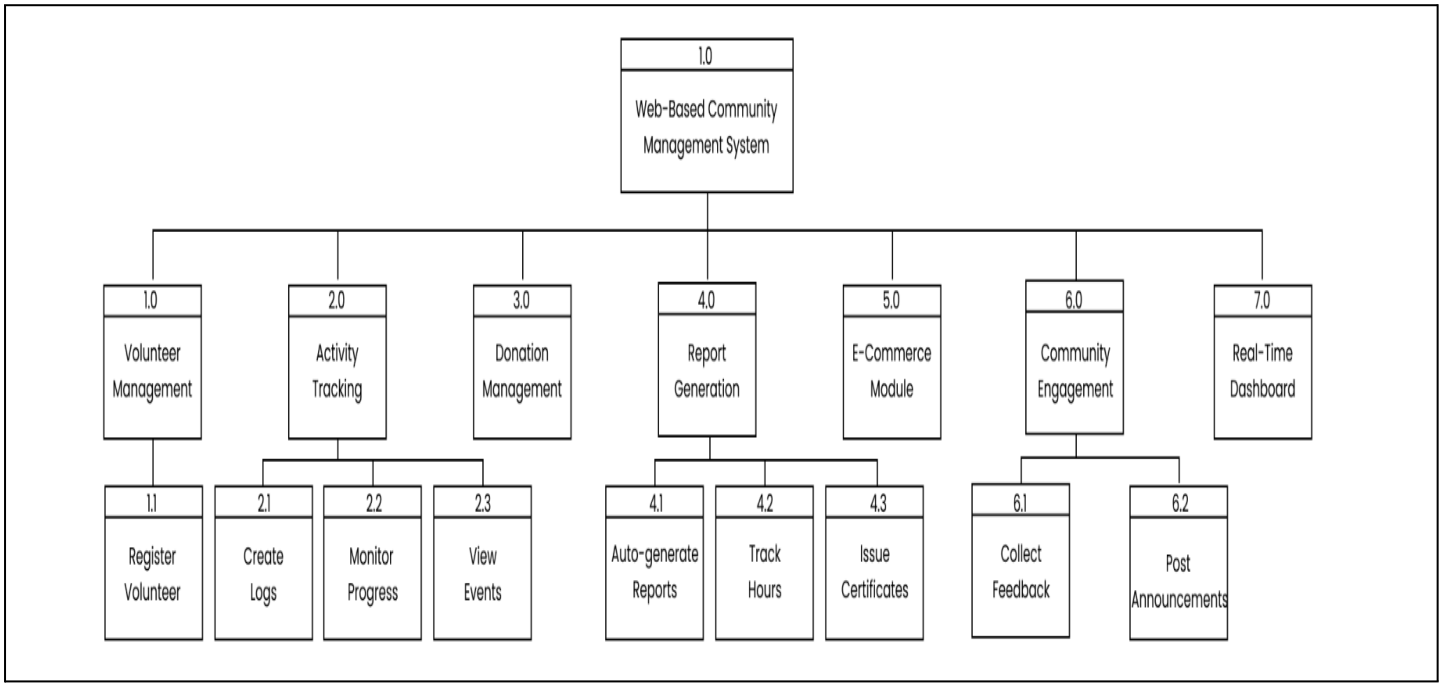


Figure 5 . HIPO Diagram

Figure 5 shows the HIPO diagram for the Web-Based Community Management System of the HappYness Project. It includes modules for Volunteer Management, Activity Tracking, Event Engagement, Community Feedback, Announcements, and Donation Handling. Each module handles specific functions like volunteer registration, project monitoring, event organization, feedback collection, announcements, and donation management, ensuring efficient operation and coordination.

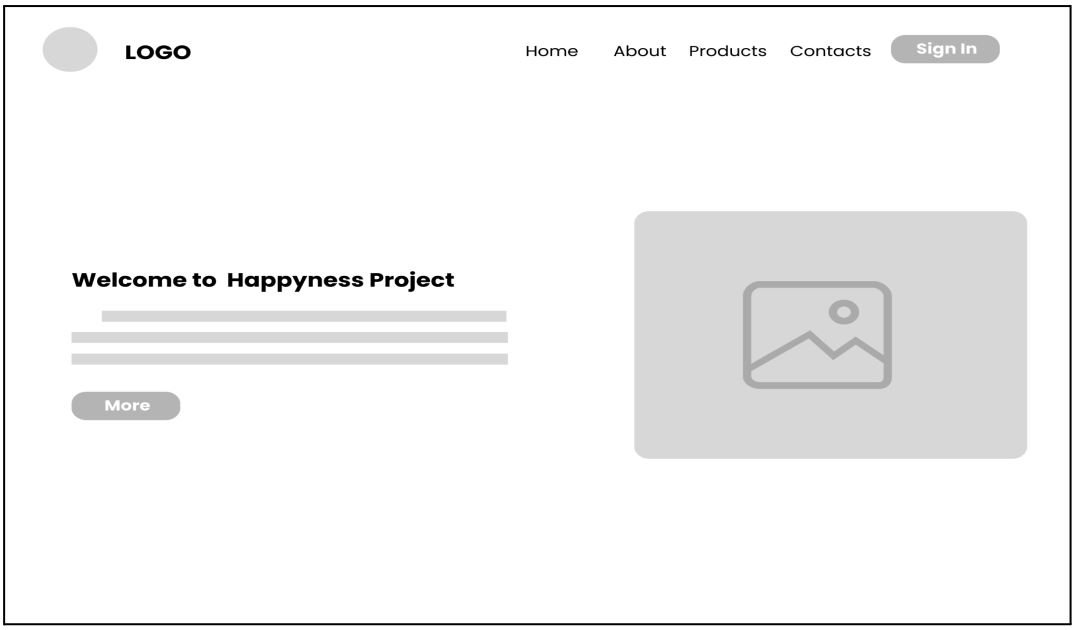


Figure 6. System Wireframe

Figure 6 shows the system wireframe of the system's landing page, showing the initial layout, navigation structure, and key interface elements such as menus, banners, and access options.

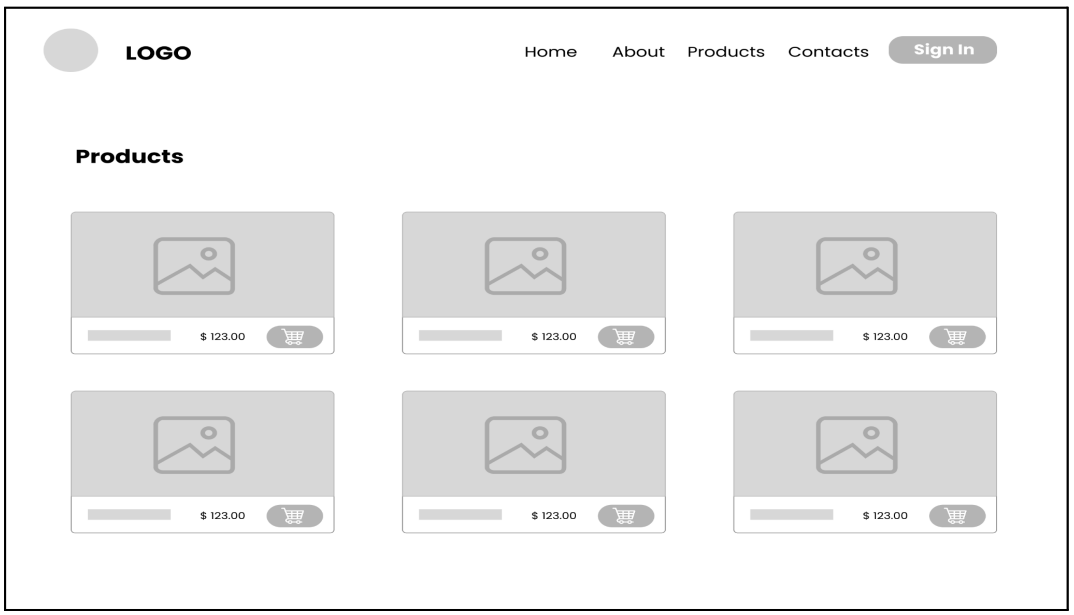


Figure 7. Product page Wireframe

Figure 7 illustrates how community-made items, descriptions, and images are organized, serving as a guide for the interface and user interaction during the design phase.

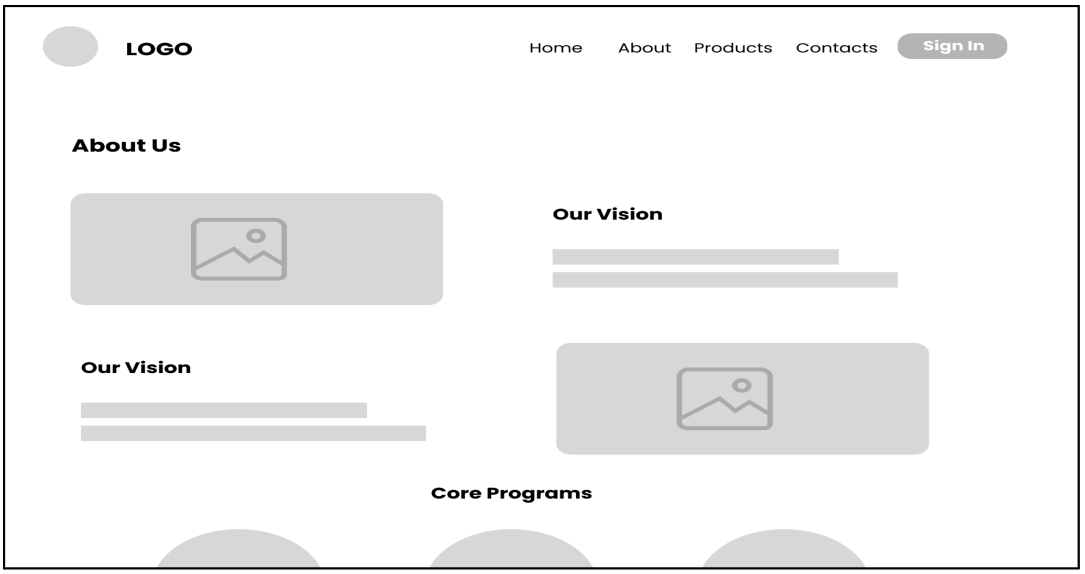


Figure 8. Wireframe of the About Us page

This diagram outlines how the HappYness Project's mission and background are organized, providing a blueprint for the site's structure and information flow during the design phase.

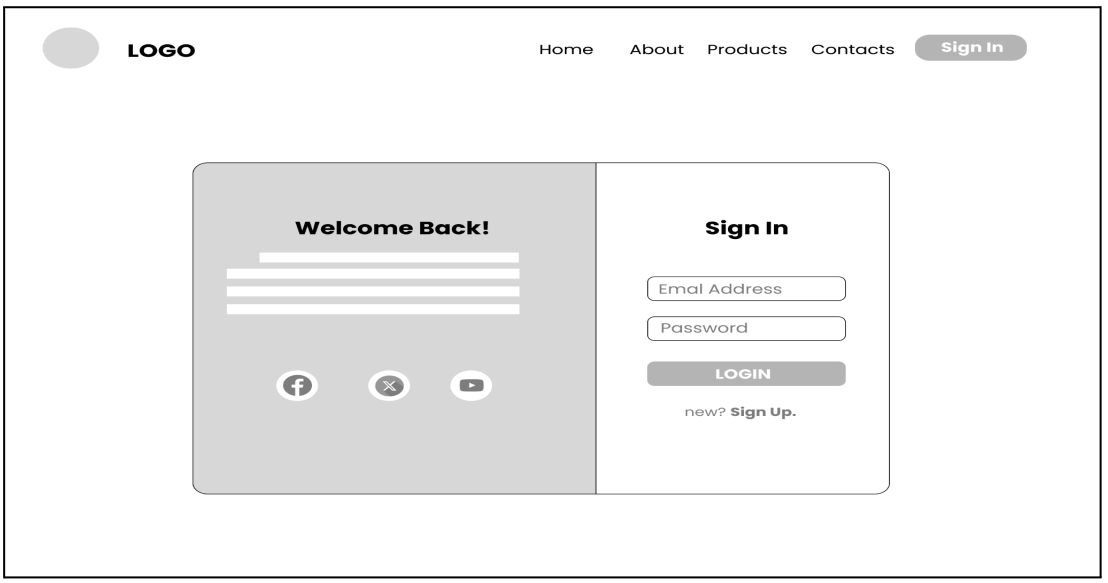


Figure 9. Login page Wireframe

Login page wireframe, showing the layout for user authentication. This includes the fields for credentials, the login button, and navigation links, serving as a basic guide for the system's entry point before development.

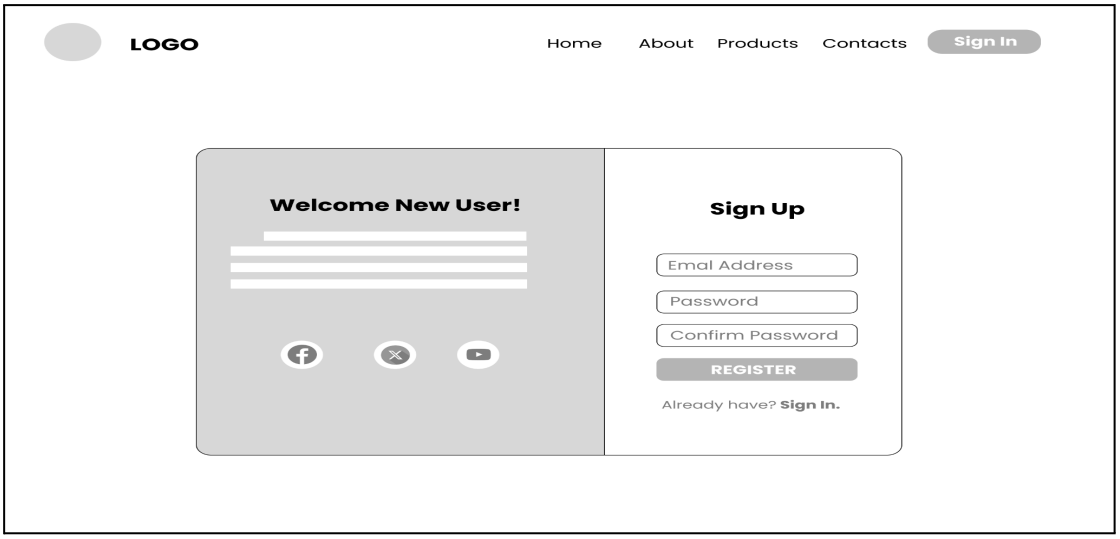


Figure 10. Wireframe for the signup page

This layout defines the placement of registration fields and account controls, providing a clear framework for the user onboarding process and data collection requirements.

Requirement Documentation

Based on the analysis of the project's objectives and discussions with stakeholders, the following features have been identified for the Web-Based Community Management System for the HappYness Project:

Table 1. Features of the Proposed System

Main Features	Detailed Features
Volunteer Management	1. Allows volunteers to register and maintain personal profiles, including assigned tasks and availability schedules. 2. Enables administrators to assign specific roles and duties to volunteers, particularly social work students (e.g., teaching, rag-making, planting). 3. Tracks OJT hours, task completion, and performance evaluations, with automated updates and report generation.
Activity Management	1. Provides guides and tutorials for activities like rag-making, planting, and teaching kids. 2. Supports event scheduling with task assignments and notifications for volunteers. 3. Stores activity data for monitoring and future planning.
Donation and Outreach Distribution	1. Maintains a beneficiary database including profiles of enrolled mothers and children. 2. Automates the scheduling and organization of outreach events with alerts for involved parties 3. Tracks donations (monetary and in-kind), including inventory management and distribution logs. 4. Documents impact through data and media uploads (e.g., goods delivered, event photos). 5. Collects feedback from beneficiaries regarding received donations and services.

Main Features	Detailed Features

Community Engagement Tools	1. Provides a built-in messaging system for secure communication between administrators, volunteers, and members. 2. Sends announcements for upcoming events, program launches, and important updates. 3. Supports community forums for open discussion and experience-sharing among members. 4. Offers special discussion sections for mothers and children to exchange support and advice. 5. Publishes success stories and testimonials to promote engagement and highlight impact.
E-Commerce and Fundraising	1. Hosts an online store showcasing handmade products created in community projects. 2. Supports multiple online payment methods including bank transfers and donation slips. 3. Provides space for promotional advertisements to generate additional program revenue.
Event Calendar	1. Displays all scheduled community events, activities, and training sessions in a centralized view. 2. Sends automated reminders and notifications to participants about upcoming events.
Reporting and Analytics	1. Generates detailed reports on volunteer activity, OJT tracking, and evaluation results. 2. Monitors key impact metrics, including children educated, rags produced, trees planted, and distributed donations. 3. Provides donor transparency through visualized reports on fund utilization and program reach

Table 2. Non-Functional Requirements of the System

Quality Attributes	**Description**
Time Behavior	Pages and features shall load within 5 seconds for optimal user experience.
Usability	The system shall offer a consistent and intuitive interface across all modules.

Accessibility	Supports multiple languages and includes features for color-blind and visually impaired users.
Security	Inactive users shall be logged out automatically; all sensitive data shall be encrypted.
Scalability	Capable of supporting up to 1,000 concurrent users without performance issues.
Compliance	Adheres to WCAG 2.1 standards to ensure accessibility for all users.

Requirements Analysis

The researcher collected comprehensive information about the study's needs and requirements during the requirements analysis phase. This was done by conducting interviews

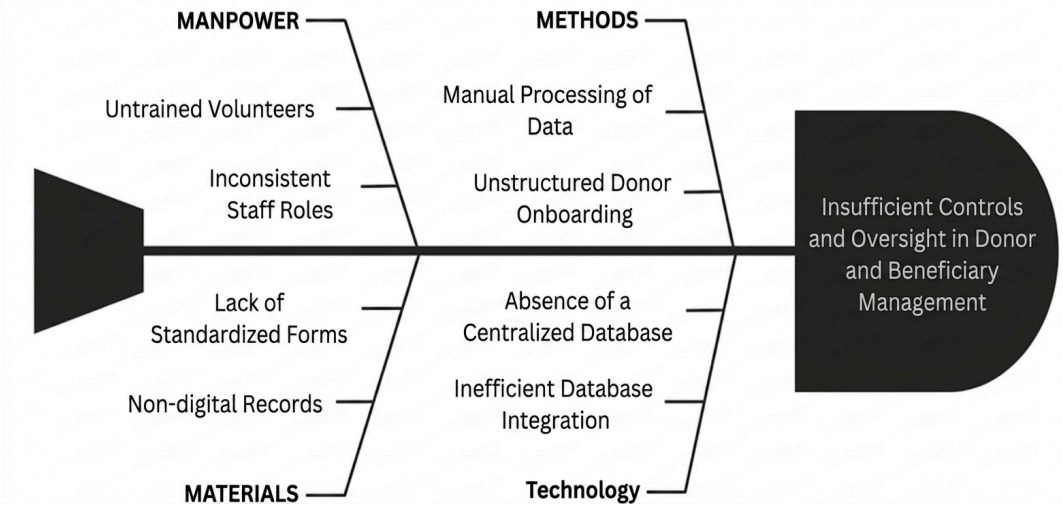


Figure 11. **Fishbone Diagram: Issues in Donor and Beneficiary Management**

Shows the Issues in Donor and Beneficiary Management by categorizing them into manpower, methods, materials, and technology, highlighting factors such as limited training, manual processes, lack of standardized forms, and the absence of a centralized database.

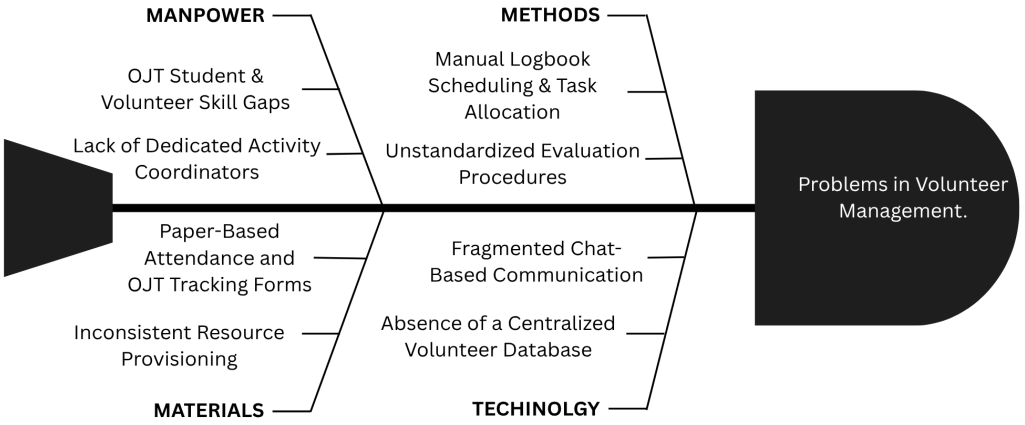
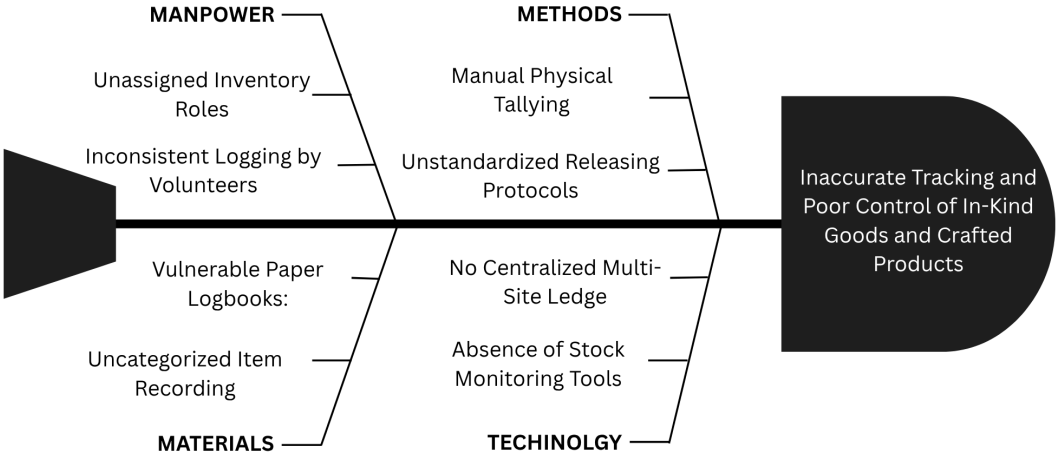


Figure 12. Fishbone Diagram: Problems in Volunteer Management

Shows the Problems in Volunteer Management by categorizing them into manpower, methods, materials, and technology, highlighting factors such as volunteer skill gaps, manual scheduling, paper-based tracking, and the absence of a centralized volunteer database.



Fishbone 13. Fishbone Diagram: Factors Affecting Inventory and Resource Management

Shows the Factors Affecting Inventory and Resource Management by categorizing them into manpower, methods, materials, and technology,

highlighting factors such as the lack of inventory staff, manual tallying, paper logbooks, and the absence of a centralized stock ledger.

System Development

The development of the Web-Based Community Management System for the HappYness Project followed the Scrum-Agile framework, an iterative and incremental approach that allowed the research team to adapt to changing requirements and deliver functional increments of the system over time. Scrum breaks down the project lifecycle into structured development cycles known as Sprints, supported by defined artifacts and ceremonies to ensure continuous stakeholder engagement and rapid progress tracking.

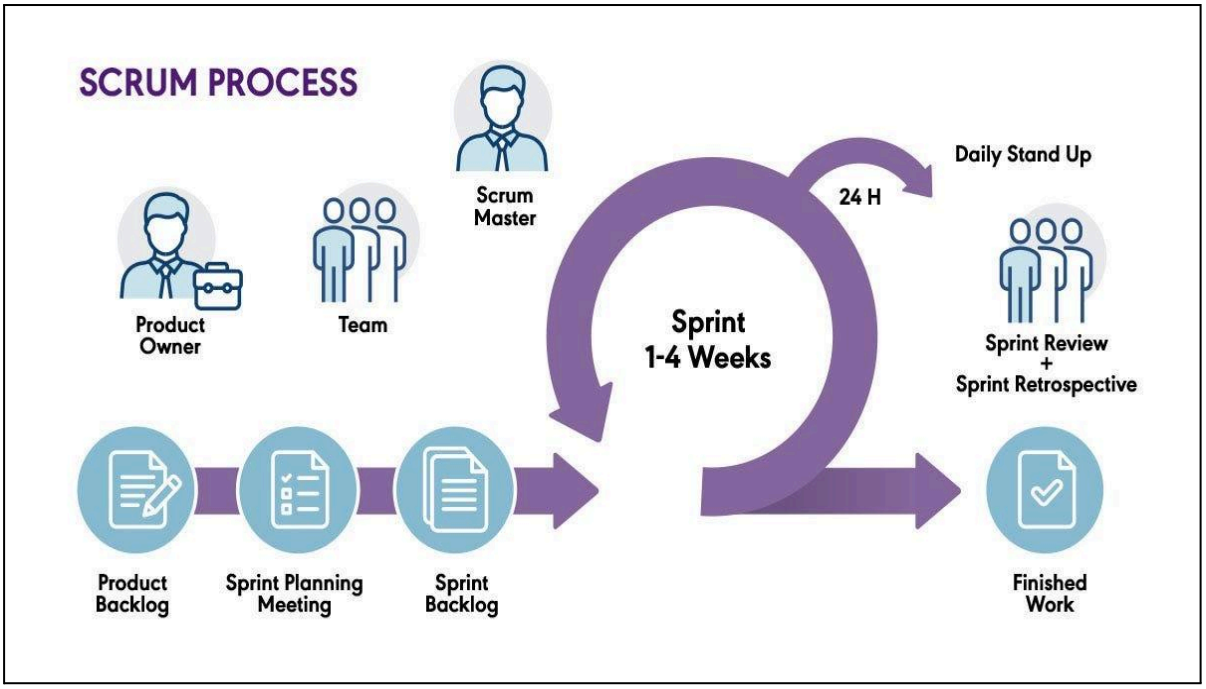


Figure 14. Scrum-Agile Development Frame Work

Requirements. The foundation of the Scrum process began with the creation of the Product Backlog, which served as the centralized, prioritized repository of all project requirements, features, and fixes. The Requirements phase

involved the initial stage of identifying and documenting the project's goals, scope, and specific software needs through interviews and the analysis of current manual processes of the HappYness Project. The team identified functional requirements (such as specific features for volunteer registration, donation logging, and automated certificate generation for social work students) and non-functional requirements (system quality and performance standards). Furthermore, the technical roadmap was established, selecting a web-based architecture using HTML5, CSS3, and JavaScript for the front-end, and PHP/Python with SQLite3/MySQL for the back-end to ensure accessibility across project sites in Bacoor, Quiapo, and Binangonan.

Design. At the start of each development cycle, the team conducted Sprint Planning sessions to select a set of high-priority items from the Product Backlog to be completed in the upcoming Sprint. During this phase, the research team created initial design concepts, including wireframes (Figures 6–10) and Use Case Diagrams, reflecting the features required by stakeholders. The system architecture was structured to support different user roles—Admin, Volunteer, and Beneficiary—ensuring that the interface and functionalities aligned with their respective responsibilities. Through stakeholder consultations and collaborative refinements, the system's visual structure and user flows were finalized.

Development. During the Sprint execution, the team focused on the actual construction of the system modules using modern web technologies (HTML5, CSS3, JavaScript for the frontend, and PHP 8.x with MySQL 8.0 for backend development and database management). The system was divided into modular increments: Volunteer Management, Donation Tracking, Activity Monitoring, Inventory Management, E-Commerce, and Reporting. Throughout

this phase, the team maintained alignment with sprint goals while developing features incrementally. *Note:* Daily short alignment checks (Daily Scrums) were practiced informally by the team to synchronize tasks and address roadblocks immediately.

Testing. Once a functional increment or module was developed within a Sprint, it underwent rigorous testing. Each feature was tested for functionality, data accuracy, and user experience. Bugs and logical issues identified during this phase were documented as backlog items and resolved promptly. The goal of this phase was to ensure the stability and reliability of the system and to verify that it fulfilled the operational needs outlined by the HappYness Project stakeholders.

Deployment. After successful testing and completion of sprint deliverables, functional increments of the system were deployed in a controlled environment for real-world use. This involved installing the web application on a live server and configuring it for user access. Admin users were trained on how to navigate and manage the platform, and initial deployment documentation (live server files, configuration files, user manuals, and training documentation) was finalized. The action performed in this phase is the release and configuration of the system in a live environment for end-user access and operational use.

Review. Each cycle concluded with a Sprint Review (and feedback evaluation) where system performance was reviewed and feedback was gathered from users and project beneficiaries. The review process helped identify areas for improvement and additional feature requests, which were fed back into the Product Backlog as new user stories or refinement tasks. Additionally, a retrospective approach was utilized to evaluate team processes, ensuring that

future iterations of the Agile cycle better supported the evolving needs of the HappYness Project.

Data Analysis Plan

This section explains how the data gathered from the evaluation of the Web-Based Community Management System for the HappYness Project will be organized, processed, and interpreted. Table 3 shows the breakdown of the respondents who will evaluate the system. The evaluation will use a questionnaire based on the ISO 25010 software quality model. Responses will be encoded, tallied per indicator, and summarized using the percentage and the weighted mean. The results will then be given a verbal interpretation using the five-point Likert scale in Table 4.

Table 3. Breakdown of Respondents for the Evaluation

CLASSIFICATIONS	FREQUENCY	PERCENTAGE
Beneficiaries	**30**	**62.50%**
Admin/Staff	**8**	**16.67%**
IT Professionals	**10**	**20.83%**
Total	**48**	**100.00%**

Implementation Plan

The implementation of the Web-Based Community Management System for the HappYness Project will follow a clear and organized plan to ensure its success. First, the researchers will prepare by confirming the project's goals and scope and by identifying the needed resources, such as development tools, hosting platform, and personnel. The researchers will work

closely with the administrators, staff, volunteers, and the client organization to present and demonstrate the system's main features for each type of user, making sure the functions meet their needs. Afterward, the system will be installed and configured on a secure web hosting platform so it can be accessed from Bacoor, Quiapo, and Binangonan, and its settings will be adjusted to meet the specific requirements of the HappYness Project. Existing records, such as volunteer lists, donation logs, and beneficiary profiles, will then be encoded into the system, and the staff will be trained to use and maintain it. Finally, a user manual and maintenance guidelines will be given to the organization to support the long-term use of the system.

Statistical Treatment of Data

1. Weighted Mean

The weighted mean was utilized to determine the average evaluation rating given by the different respondent groups for each indicator under the various ISO 25010 quality characteristics. This method assigns appropriate weights to each scale category to reflect its relative contribution to the overall evaluation.

The formula for the weighted mean is:

$$\mu = \frac{\Sigma f_i x_i}{N}$$

Where:

μ = Weighted mean

$\sum$ = Summation of

f_i = Frequency

x_i = Weight

N = Total number of respondents (48)

2. Five-Point Likert Scale

A five-point Likert scale was used in the evaluation questionnaires to measure the degree of agreement and performance of the system. The computed weighted means were interpreted using the following statistical limits and verbal equivalents:

Table 2. The Five-Point Likert Scale Used in Rating the Software

RATE	EQUIVALENT	INTERVAL INTERPRETATION
5	**4.50 – 5.00**	**Excellent**
4	**3.50 – 4.49**	**Very Good**
3	**2.50 – 3.49**	**Good**
2	**1.50 – 2.49**	**Fair**
1	**1.00 – 1.49**	**Poor**

RESULTS AND DISCUSSION

This chapter presents the description of the developed system, how it was developed and tested, and the results of the evaluation done by the respondents.

SYSTEM DESIGN

The Web-Based Community Management System for the HappYness Project was developed to help the organization manage its volunteers, activities, donations, and beneficiaries in one place. Before the system, these records were kept on paper logbooks and chat groups, which caused delays and inconsistent records. The system has four types of users: the Regular User, the Admin, the Volunteer, and the Beneficiary. Each user type can only see and use the pages that match their role.

The main modules of the system are Volunteer Management, Activity Management, Donation Management, Beneficiary Management, Report Generation, E-Commerce, Community Engagement, and the Real-Time Dashboard. The figures below show the main pages of the system.

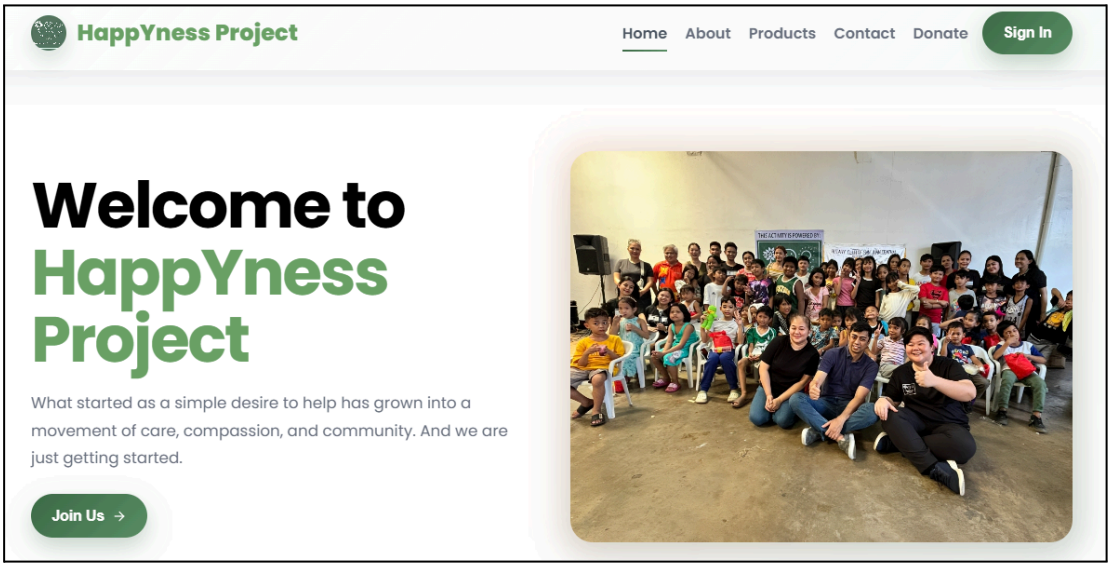


Figure 16. Homepage

Figure 16 shows the homepage of the system. It gives a short introduction to the HappYness Project and links to the login and sign-up pages, the product page, and the About Us page.

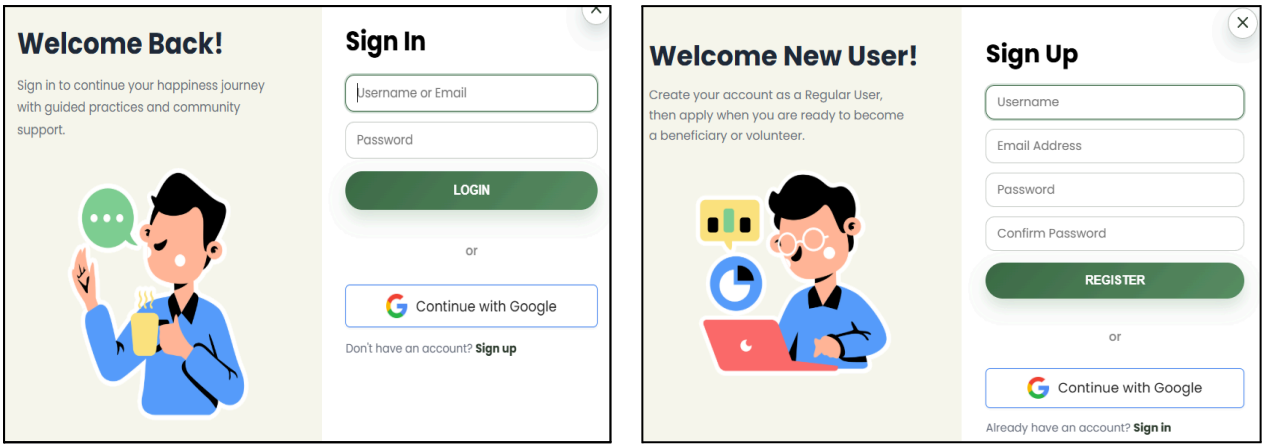


Figure 17. Login and Sign-up Pages

Figure 17 shows the login and sign-up pages. Users must have an account to enter the system. After logging in, the user is taken to the pages that match the user's role.

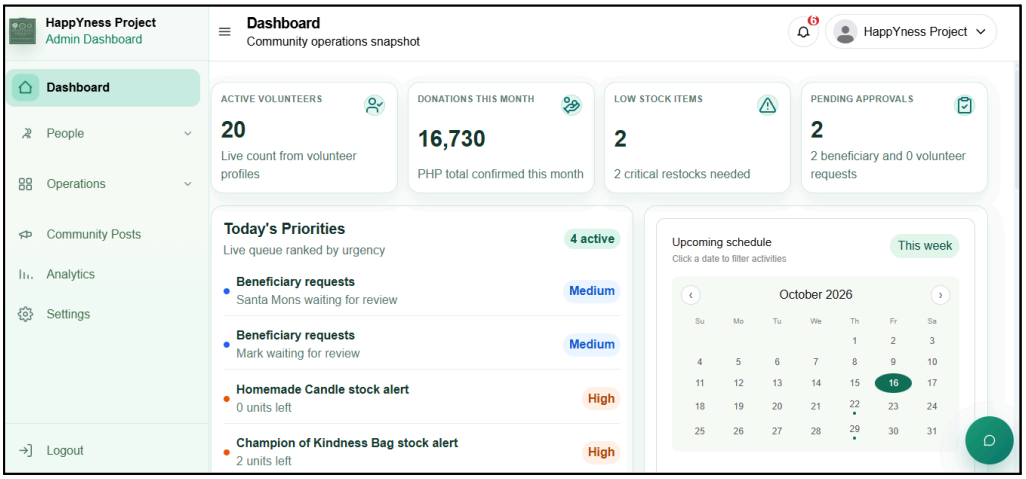


Figure 18. Admin Dashboard

Figure 18 shows the Admin Dashboard. It shows the latest data on donations, volunteer participation, and the status of activities, so the administrators no longer need to count and encode these by hand.

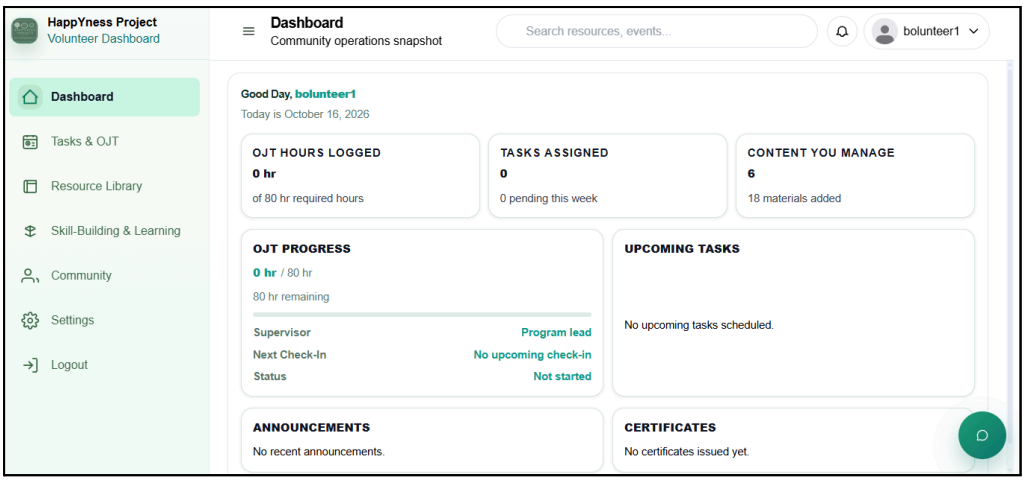


Figure 19. Volunteer Dashboard

Figure 19 shows the Volunteer Management page. The administrators can register volunteers, approve them, and assign tasks such as teaching, rag-making, and planting. Volunteers can also view their assigned activities and their participation history.

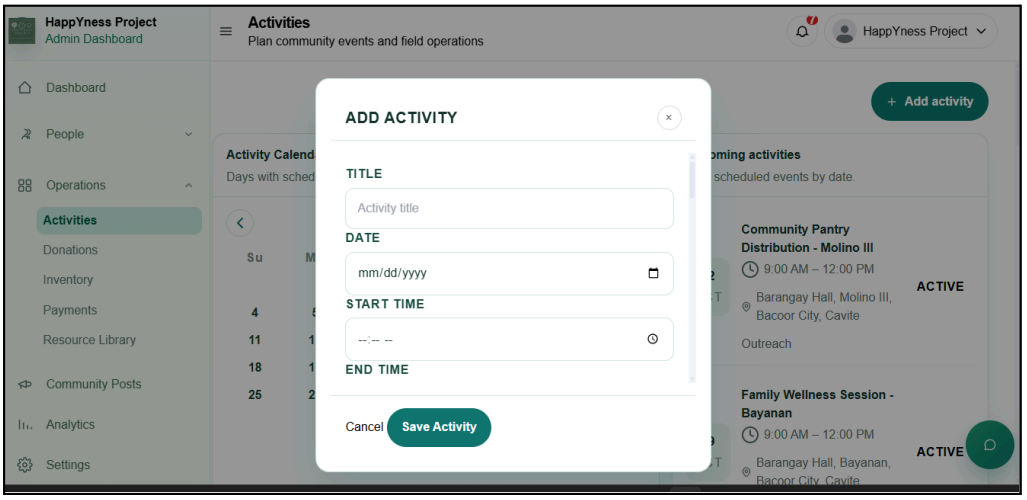


Figure 20. Activity Management Page

Figure 20 shows the Activity Management page. It is used to create activities, set their schedule, and update their status. Volunteers can record their participation in the activities assigned to them.

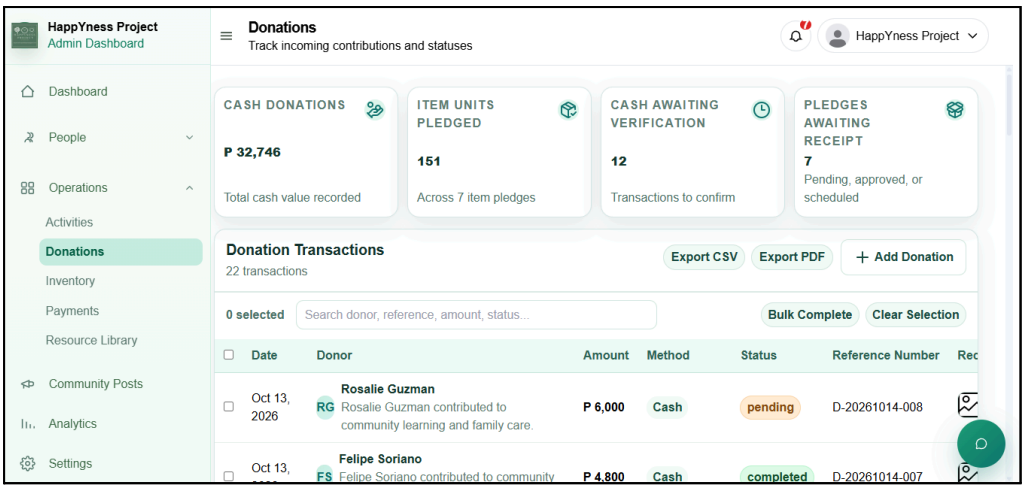


Figure 21. Donation Management Page

Figure 21 shows the Donation Management page. It records the donations received, whether money or goods, and the donations given out to the communities. This keeps the inventory records accurate and easy to check.

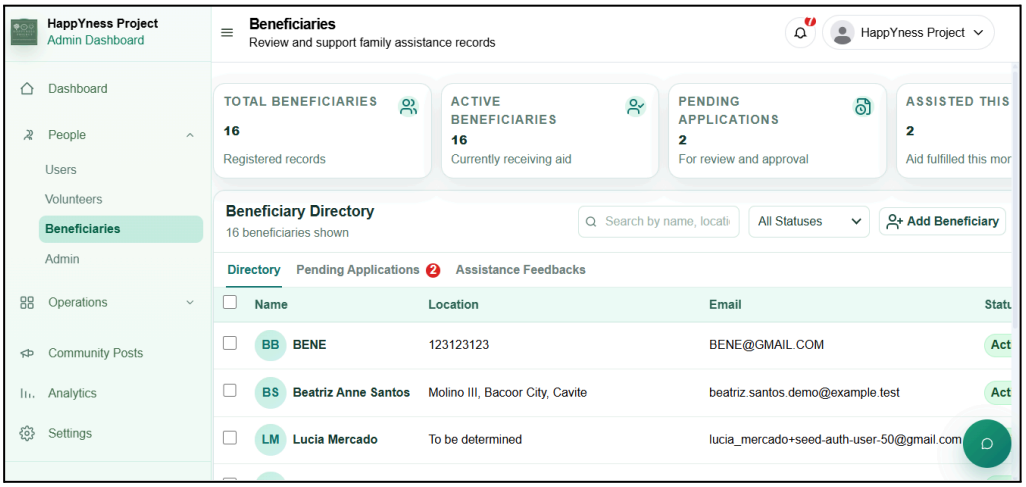


Figure 22. Beneficiary Management Page

Figure 22 shows the Beneficiary Management page, where the administrators register and verify beneficiaries. Beneficiaries can update their information and view the programs they joined.

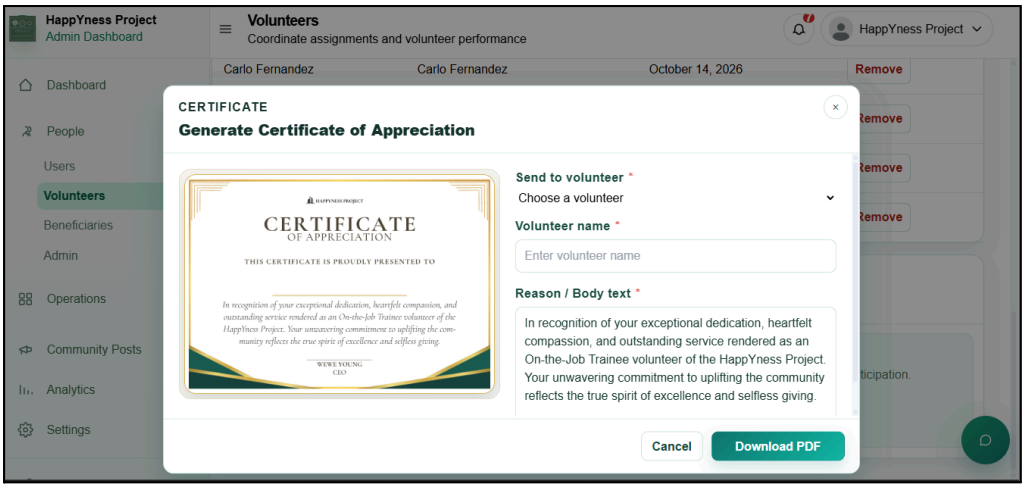


Figure 23. Reports and Certificates

Figure 23 shows the Report Generation feature. The system compiles the data from the other modules into reports that can be printed or saved as PDF. It also supports the generation of participation certificates for volunteers.

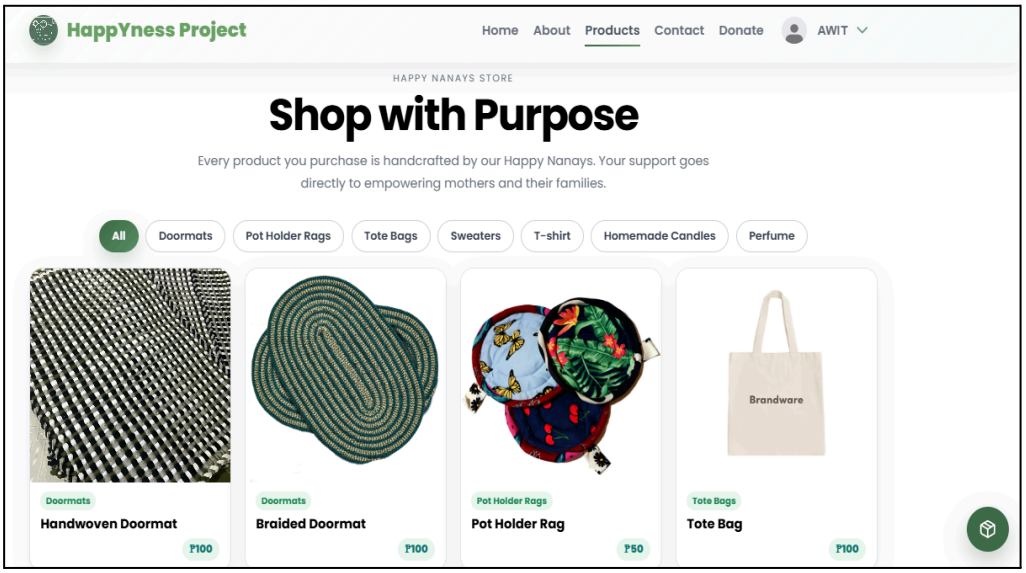


Figure 24. Online Store

Figure 24 shows the online store, where the products made by the community, such as rags, plants, and crafts, are listed and sold to support the project.

System Development

The Web-Based Community Management System for the HappYness Project was developed using the Agile model. This approach allowed the researchers to improve the system through repeated cycles of requirements gathering, design, development, testing, and review. During the requirements phase, the researchers interviewed the HappYness Project staff and studied their manual processes to identify what the organization needed to manage its volunteers, activities, donations, and beneficiaries. In the design phase, the use case diagram, HIPO diagram, and wireframes were prepared based on these requirements. The development phase involved building the system using HTML5, CSS3, and JavaScript (ES6) for the user interface, Python 3.10 with Django for the server-side processes, and SQLite3 for data storage, followed by unit testing, integration testing, and system testing to check that the features work correctly. The HappYness Project staff reviewed the working modules after each cycle, and their comments were used to improve the system. After testing and revisions, the system was deployed to use by the organization.

System testing

System testing for the Web-Based Community Management System for the HappYness Project was conducted to verify that all integrated modules—including volunteer management, activity tracking, donation control, beneficiary administration, and the e-commerce store—performed reliably and met the specified functional requirements. The testing process evaluated the system across multiple levels, beginning with unit testing to validate individual components, followed by integration testing to ensure seamless data communication between the front-end interface, Python/Django back-end logic, and the SQLite3 database. Further tests were executed to assess performance efficiency, time behavior, security protocols, and usability under

various user roles (Regular user, Admin, Volunteer, and Beneficiary). The results confirmed that the application effectively eliminated manual data-processing bottlenecks, operated stably under expected workloads, and fulfilled the quality criteria necessary for deployment.

Table 4. Software test environment results under functionality criteria

CRITERIA	FREQUENCY	PERCENT
Admin Account Module		
1. The administrator was able to securely log in using authorized credentials.	4	100%
2. The administrator was able to manage user accounts (approve, edit, or deactivate regular users, volunteers, and beneficiaries).	4	100%
3. The administrator was able to create, schedule, and assign community activities (teaching, planting, rag-making).	4	100%
4. The administrator was able to track and log incoming monetary and in-kind donations as well as inventory levels.	4	100%
5. The administrator was able to view real-time dashboards and generate automated PDF reports.	4	100%
Volunteer Account Module		
1. The volunteer was able to register, log in, and maintain a personal profile with availability schedules.	4	100%

2. The volunteer was able to view assigned tasks, sign up for upcoming outreach events, and receive notifications.	4	100%

Table.4 Continue

CRITERIA	FREQUENCY	PERCENT
3. The system accurately tracked OJT hours, task completion status, and performance logs.	4	100%
4. The volunteer was able to download automated digital participation certificates upon task completion.	4	100%
Beneficiary Account Module		
1. The beneficiary was able to view verified profiles and access community support programs.	4	100%
2. The beneficiary was able to apply for assistance and view records of Distributed goods or services received.	4	100%
3. The beneficiary was able to access educational tutorials (e.g., basic child care, parenting articles).	4	100%
4. The beneficiary was able to submit feedback regarding received donations and community services.	4	100%
Regular User Account Module		
1. The regular user was able to create	4	100%

an account, log in, and reset passwords successfully.		
2. The regular user was able to browse the e-commerce catalog displaying community	4	100%

Table.4 Continue

CRITERIA	FREQUENCY	PERCENT
-made products (rags, crafts).		
3. The regular user was able to initiate product purchases and view transaction details.	4	100%
4. The regular user was able to view public announcements, event calendars, and general project updates.	4	100%

System Evaluation

The performance of the Web-Based Community Management System for the HappYness Project was evaluated based on the ISO 25010 software quality model. This includes functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. There were 48 respondents, including 30 beneficiaries, 8 admin/staff, and 10 IT professionals, who provided feedback through a structured survey to assess the application's quality and overall effectiveness in meeting user needs.

In Table 2, the evaluation results for the functional suitability assessment criteria of the application are presented. Under this assessment, functional completeness achieved a weighted mean of 4.29, corresponding to a verbal interpretation of "Very Good." Similarly, functional correctness

attained a weighted mean of 4.19, receiving a "Very Good" rating. Lastly, functional appropriateness obtained a weighted mean of 4.17, indicating a "Very Good" level. Overall, the functional suitability category achieved a general weighted mean of 4.22, corresponding to a verbal interpretation of "Very Good."

Table 2. Functional suitability assessment of the software

FUNCTIONAL SUITABILITY	W.M.	VERBAL INTERPRETATION
1. Functional completeness - if the functions cover all the specified tasks and user objectives.	4.27	Very Good
2. Functional correctness - if the system provides the correct results with needed degree of precision.	4.21	Very Good
3. Functional appropriateness - if the functions facilitate the accomplishment of specified tasks and objectives.	4.17	Very Good
GENERAL WEIGHTED MEAN	**4.22**	**VERY GOOD**

In Table 3, the evaluation results for the performance efficiency assessment criteria of the application are presented. Under this assessment, time behavior achieved a weighted mean of 4.06, corresponding to a verbal interpretation of "Very Good." Similarly, resource utilization attained a weighted mean of 4.08, receiving a "Very Good" rating. Lastly, capacity obtained a weighted mean of 4.27, indicating a "Very Good" level. Overall, the

performance efficiency category achieved a general weighted mean of 4.14, corresponding to a verbal interpretation of "Very Good."

Table 3. Performance efficiency assessment of the software

FUNCTIONAL SUITABILITY	W.M.	VERBAL INTERPRETATION
1. Time behavior - if the system response and processing times and throughput rates, when performing its functions, meet requirements.	4.08	Very Good
2. Resource utilization - if the system amounts and types of resources, when performing its functions, meet requirements.	4.08	Very Good
3. Capacity - if the system maximum limits meet requirements.	4.27	Very Good
GENERAL WEIGHTED MEAN	**4.15**	**VERY GOOD**

In Table 4, the evaluation results for the compatibility assessment criteria of the application are presented. Under this assessment, coexistence achieved a weighted mean of 4.21, corresponding to a verbal interpretation of "Very Good." Similarly, interoperability attained a weighted mean of 4.02, receiving a "Very Good" rating. Overall, the compatibility category achieved a general weighted mean of 4.12, corresponding to a verbal interpretation of "Very Good."

Table 2. Compatibility assessment of the software

COMPATIBILITY	W.M.	VERBAL INTERPRETATION
1. Co - existence - if the system can perform its required functions efficiently	4.23	Very Good

Table.2 Continue

	while sharing a common environment and resources with other system, without detrimental impact on any other systems.		
2.	Interoperability - if the system can exchange information and use the information that has been exchanged.	4.02	Very Good
	GENERAL WEIGHTED MEAN	**4.14**	**VERY GOOD**

In Table 5, the evaluation results for the usability assessment criteria of the application are presented. Under this assessment, appropriateness recognizability achieved a weighted mean of 4.15, learnability attained 4.06, operability obtained 4.19, user error protection recorded 4.00, user interface aesthetics reached 4.13, and accessibility secured 4.25, all corresponding to a verbal interpretation of "Very Good." Overall, the usability category achieved a general weighted mean of 4.13, corresponding to a verbal interpretation of "Very Good."

Table 5. Usability assessment of the software

	FUNCTIONAL SUITABILITY	**W.M.**	**VERBAL INTERPRETATION**
1.	Appropriateness recognizability - if the system is appropriate for the need of the users.	4.15	Very Good
2.	Learnability - if the system can be used by specified users to achieve specified goals in	4.04	Very Good

Table.5 Continue

learning to use the system with effectiveness, efficiency, freedom, and control.		
3. Operability - if the system has attributes that make it easy to operate and control.	4.21	Very Good
4. User error protection - if the system protects users against making errors.	4.00	Very Good
5. User interface aesthetics - if the system user interface enables pleasing and satisfying interaction for the user.	4.13	Very Good
6. Accessibility - if the system can be used by the people with the widest range of characteristics and capabilities to achieve specified goal in a specified context of use.	4.25	Very Good
GENERAL WEIGHTED MEAN	**4.13**	**VERY GOOD**

In Table 6, the evaluation results for the reliability assessment criteria of the application are presented. Under this assessment, maturity achieved a weighted mean of 4.09, availability attained 4.08, fault tolerance recorded 4.06, and recoverability obtained 3.92, all corresponding to a verbal interpretation of "Very Good." Overall, the reliability category achieved a general weighted mean of 4.04, corresponding to a verbal interpretation of "Very Good."

Table 6. Reliability assessment of the software

RELIABILITY	W.M.	VERBAL INTERPRETATION
1. Maturity - if the system meets need for reliability under normal operation.	4.09	Very Good
2. Availability - if the system is operational and accessible when required for use.	4.08	Very Good
3. Fault tolerance - if the system operates as intended despite the presence of hardware or software faults.	4.04	Very Good
4. Recoverability - if the system, in the event of an interruption or a failure, can recover the data directly affected and re-establish the desired state of the system.	3.92	Very Good
GENERAL WEIGHTED MEAN	**4.03**	**VERY GOOD**

In Table 7, the evaluation results for the security assessment criteria of the application are presented. Under this assessment, confidentiality achieved a weighted mean of 4.06, integrity attained 4.02, non-repudiation recorded 3.94, accountability achieved 4.17, and authenticity obtained 3.98, all corresponding to a verbal interpretation of "Very Good." Overall, the security category achieved a general weighted mean of 4.03, corresponding to a verbal interpretation of "Very Good."

Table 7. Security assessment of the software

RELIABILITY	W.M.	VERBAL INTERPRETATION
1. Confidentiality - if the system ensures that the data are accessible only to those authorized to have access.	4.08	Very Good
2. Integrity - if the system prevents unauthorized access to, or modification of, computer programs or data.	4.04	Very Good
3. Non-repudiation - if the system's action or events can be proven to have taken place so that the events or action cannot be repudiated later.	3.94	Very Good
4. Accountability - if the system's actions of an entity can be traced uniquely to the entity.	4.17	Very Good
5. Authenticity - if the system's identity of a subject or resource can be proved to be the one claimed.	4.00	Very Good
GENERAL WEIGHTED MEAN	**4.04**	**VERY GOOD**

In Table 8, the evaluation results for the maintainability assessment criteria of the application are presented. Under this assessment, modularity achieved a weighted mean of 4.27, reusability attained 4.27, analyzability recorded 4.08, modifiability achieved 4.04, and testability obtained 4.02, all corresponding to a verbal interpretation of "Very Good." Overall, the

maintainability category achieved a general weighted mean of 4.14, corresponding to a verbal interpretation of "Very Good."

Table 8. Maintainability assessment of the software

MAINTAINABILITY	W.M.	VERBAL INTERPRETATION
1. Modularity - if the system composed of discrete components such that a change to one component has minimal impact on other components.	4.27	Very Good
2. Modularity - if the system composed of discrete components such that a change to one component has minimal impact on other components.	4.27	Very Good
3. Analyzability - if the system is possible to assess the impact of an intended change to one or more of its parts, or to diagnose the system for deficiencies or causes of failures, or to identify parts to be modified.	4.08	Very Good
4. Modifiability - if the system can be effectively and efficiently modified without introducing defects or degrading existing product quality.	4.04	Very Good
5. Testability - if the system can establish test criteria and	4.02	Very Good

Table.8 Continue

can be performed to determine whether those criteria have been met.		
GENERAL WEIGHTED MEAN	**4.14**	**VERY GOOD**

In Table 9, the evaluation results for the portability assessment criteria of the application are presented. Under this assessment, adaptability achieved a weighted mean of 3.96, installability attained 3.98, and replaceability obtained 3.90, all corresponding to a verbal interpretation of "Very Good." Overall, the portability category achieved a general weighted mean of 3.94, corresponding to a verbal interpretation of "Very Good."

Table 9. Portability assessment of the software

FUNCTIONAL SUITABILITY	**W.M.**	**VERBAL INTERPRETATION**
1. Adaptability - if the system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.	3.98	Very Good
2. Instability - if the system can be successfully installed and/or uninstalled in a specified environment.	4.00	Very Good
3. Replaceability - if the system can replace another specified software product for the same purpose in the same environment.	3.88	Very Good
GENERAL WEIGHTED MEAN	**3.95**	**VERY GOOD**

In Table 10, the summary of the software quality assessment is presented. Functional Suitability garnered the highest general weighted mean at 4.22, followed by Performance Efficiency (4.14), Maintainability (4.14), Usability (4.13), Compatibility (4.12), Reliability (4.04), Security (4.03), and Portability (3.94). The total overall rating for the Web-Based Community Management System is 4.09, corresponding to a verbal interpretation of "Very Good," validating the overall technical success, high quality, and readiness of the platform for the HappYness Project.

Table 10. Summary table showing the assessment of the software

EVALUATION CRITERIA	W.M.	VERBAL INTERPRETATION
1. Functional Suitability	4.22	Very Good
2. Performance Efficiency	4.15	Very Good
3. Compatibility	4.14	Very Good
4. Usability	4.13	Very Good
5. Reliability	4.03	Very Good
6. Security	4.05	Very Good
7. Maintainability	4.14	Very Good
8. Portability	3.95	Very Good
TOTAL RATING	**4.10**	**VERY GOOD**

Table 11 presents a comparative analysis of the perceptions of Beneficiaries, Admin/Staff, and IT Professionals across the different software evaluation criteria. In terms of Functional Suitability, Beneficiaries gave the highest general weighted mean (G.W.M.) of 4.34, followed by IT Professionals

at 4.13 and Admin/Staff at 3.83. For Performance Efficiency, Beneficiaries again gave the highest G.W.M. of 4.19, followed by Admin/Staff at 4.13 and IT Professionals at 4.00. Under Compatibility, Beneficiaries rated the application at 4.25, while IT Professionals and Admin/Staff gave G.W.M. ratings of 4.00 and 3.75, respectively. Regarding Usability, Beneficiaries provided the highest G.W.M. of 4.20, followed by Admin/Staff at 4.17 and IT Professionals at 3.88.

In terms of Reliability, Beneficiaries gave the highest G.W.M. of 4.14, followed by Admin/Staff at 3.91 and IT Professionals at 3.85. For Security, Beneficiaries rated the application at 4.15, followed by IT Professionals at 4.02 and Admin/Staff at 3.60. In Maintainability, Beneficiaries again gave the highest G.W.M. of 4.26, followed by IT Professionals at 3.98 and Admin/Staff at 3.88. Lastly, in Portability, Beneficiaries gave the highest G.W.M. of 4.01, followed by IT Professionals at 3.87 and Admin/Staff at 3.79.

Overall, the total average rating was highest among Beneficiaries at 4.19, followed by IT Professionals at 3.97 and Admin/Staff at 3.88. This indicates that the Beneficiaries had the most favorable overall perception of the application, while Admin/Staff rated Security the lowest at 3.60. All ratings from the three groups fall within the “Very Good” range. The overall G.W.M. of the application, as reported in Table 10, was 4.09, corresponding to a verbal interpretation of “Very Good.”

Table 11. Summary table showing the assessment of the software

EVALUATION CRITERIA	TYPE OF REPORT		
	Beneficiary	**Admin**	**IT Professional**
FUNCTIONAL SUITABILITY			
Functional completeness	4.43	3.88	4.10
Functional correctness	4.30	3.88	4.20
Functional appropriateness	4.30	3.75	4.10
G.W.M.	**4.34**	**3.83**	**4.13**
PERFORMANCE EFFICIENCY			
Time behavior	4.03	4.13	4.20
Resource utilization	4.10	4.13	4.00
Capacity	4.43	4.13	3.90
G.W.M.	**4.19**	**4.13**	**4.03**
COMPATIBILITY			
Co - existence	4.33	4.00	4.10
Interoperability	4.17	3.50	4.10
G.W.M	**4.25**	**3.75**	**4.10**
USABILITY			
Appropriateness	4.13	4.50	3.90
recognizability			
Learnability	4.03	4.25	3.90
Operability	4.27	4.25	4.00
User error protection	4.17	3.75	3.70
User interface aesthetics	4.17	4.25	3.90
Accessibility	4.43	4.00	3.90
G.W.M.	**4.20**	**4.17**	**3.88**
RELIABILITY			
Maturity	4.21	3.88	3.90
Availability	4.17	3.88	4.00
Fault tolerance	4.03	4.38	3.80
Recoverability	4.13	3.50	3.60
G.W.M	**4.14**	**3.91**	**3.83**
SECURITY			
Confidentiality	4.20	3.63	4.10
Integrity	4.17	3.50	4.10
Non-repudiation	4.00	3.63	4.00
Accountability	4.33	3.63	4.10
Authenticity	4.07	3.63	4.00
G.W.M	**4.15**	**3.60**	**4.08**
MAINTAINABILITY			
Modularity	4.47	3.75	4.10

Table. 11 Continue

Reusability	4.37	4.13	4.10
Analyzability	4.20	3.75	4.00
Modifiability	4.13	3.75	4.00
Testability	4.13	4.00	3.70
G.W.M.	**4.26**	**3.88**	**3.98**
PORTABILITY			
Adaptability	4.00	4.00	3.90
Instability	4.00	3.88	4.10
Replaceability	4.03	3.50	3.70
G.W.M.	**4.01**	**3.79**	**3.90**
TOTAL RATING	**4.19**	**3.88**	**3.99**

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter presents the summary of findings, the conclusions, and the recommendations based on the results of the testing and evaluation of the system.

Summary

The study was conducted to develop a Web-Based Community Management System for the HappYness Project. It was designed to help the organization manage its volunteers, activities, donations, and beneficiaries in one place, since these records were previously kept in paper logbooks and chat groups. The system has four types of users: the Regular User, the Admin, the Volunteer, and the Beneficiary. The Volunteer Management module is used to register volunteers and assign them to tasks. The Activity Management module is used to schedule activities and record participation. The Donation Management and Beneficiary Management modules keep the records of donations received and given, and the profiles of the beneficiaries. The system also has Report Generation, E-Commerce, Community Engagement, and a Real-Time Dashboard for the administrators.

The researchers used the Agile Model in developing the system, which involved repeated planning, design, development, testing, and review. [Write the programming language, framework, and database actually used, and make sure it matches Chapter 3.] Interviews and observation of the HappYness Project's manual process were used to identify the needs of the users. Unit, integration, and system testing were conducted to check the features and how they worked together.

The system was evaluated using an adapted ISO 25010 instrument by 48 respondents, consisting of 30 beneficiaries, 8 admin/staff, and 10 IT professionals. The system received a total rating of 4.09, interpreted as "Very Good." Functional Suitability got the highest rating at 4.22, followed by Performance Efficiency and Maintainability at 4.14, Usability at 4.13, Compatibility at 4.12, Reliability at 4.04, and Security at 4.03. Portability got the lowest rating at 3.94. All eight quality characteristics were interpreted as "Very Good."

By group, the beneficiaries gave the highest total rating of 4.19, followed by the IT professionals at 3.97 and the admin/staff at 3.88. The lowest ratings were given by the admin/staff for interoperability, integrity, recoverability, and replaceability at 3.50 each, and by the IT professionals for recoverability at 3.60.

Conclusion

The study aimed to develop a system that would help the HappYness Project manage its volunteers, activities, donations, and beneficiaries through a single web-based platform. The developed system included volunteer, activity, donation, and beneficiary management, report generation, an online store, community engagement features, and a real-time dashboard. The administrators were able to view the data and generate reports without relying on paper records. The functionality tests conducted on the system showed that its main features worked as expected, and the problems found during testing were fixed before the evaluation.

The quality of the system was supported by the evaluation of the beneficiaries, admin/staff, and IT professionals. Their total ratings were 4.19,

3.88, and 3.97, respectively. The beneficiaries gave the most favorable rating, while the admin/staff gave the lowest, especially for Security at 3.60. The total rating of 4.09, interpreted as "Very Good," showed that the evaluators gave favorable ratings to the system. Functional Suitability received the highest rating, which means the evaluators found that the system can do the tasks it was made for.

Based on the testing and evaluation results, the study concluded that the Web-Based Community Management System met its stated functions and received a "Very Good" overall rating. Within the scope of the study, the system serves as a tool for the HappYness Project to organize its records and support its community programs. The lower ratings for recoverability, interoperability, and portability show the areas that still need improvement.

Recommendation

Based on the findings and conclusions of the study, the researchers offer the following recommendations for the HappYness Project and for future researchers:

1. Develop a mobile application or a mobile-friendly version of the system, with email or SMS notifications for upcoming activities, so volunteers and beneficiaries can use it more easily.
2. Explore the integration of augmented reality (AR) technology in future versions of the WBCMS, such as allowing users to view and interact with information in a more immersive way, since this feature is beyond the scope of the current study and may be considered by future researchers as the system continues to develop.

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APPENDICES

Appendix 1. Relevant Source Code

```
<!DOCTYPE html>
{% load static %}
<html lang="en">
<head>
    <!-- Page settings -->
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>{% block title %}HappYness{% endblock %}</title>
    <link rel="icon" type="image/jpeg"
        href="{% static 'media/Happyness Project.jpg' %}?v=20261006" />
    <!-- Scripts and libraries -->
    <script src="{% static 'js/avatar-fallback.js' %}"></script>
    <script src="https://cdn.tailwindcss.com"></script>
    <script
src="https://cdn.jsdelivr.net/npm/chart.js@4.4.1/dist/chart.umd.min.js"></script
>
    <!-- Fonts -->
    <link rel="preconnect" href="https://fonts.googleapis.com" />
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin />
    <link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;7
00;800&display=swap"
        rel="stylesheet" />
    <!-- Stylesheets -->
    <link rel="stylesheet"
        href="{% static 'css/index.css'
%}?v=20261015-auth-password-reset-email" />
```

Appendix 1. Continued

```
<link rel="stylesheet"
      href="{% static 'css/admin-fixes.css' %}?v=20261002-sidebar-rail" />
  <style>
    /* ---------- 1. Default text sizes, icon size and font ---------- */
    :root {
      --dash-text-xs: 0.75rem;
      --dash-text-sm: 1rem;
      --dash-text-md: 1.08rem;
      --dash-text-lg: 1.22rem;
      --dash-text-xl: 1.25rem;
      --dash-text-2xl: 1.5rem;
      --dash-text-3xl: 1.875rem;
      --dash-icon-size: 1.3rem;
      --dash-font-family: "Poppins", -apple-system, BlinkMacSystemFont,
                          "Segoe UI", Roboto, sans-serif;
    }
    #dashboard-shell,
    #dashboard-shell *,
    #dashboard-main-wrapper,
      #dashboard-main-wrapper * {
      font-family: var(--dash-font-family) !important;
    }
    /* ---------- 2. Text size choices (set in the user's settings) ---------- */
    /* The "medium" choice uses the default values above. */
    html.dashboard-font-small {
      --dash-text-xs: 0.68rem;
      --dash-text-sm: 0.9rem;
```

Appendix 1. Continued

```
--dash-text-md: 0.98rem;
        --dash-text-lg: 1.1rem;
        --dash-text-xl: 1.15rem;
        --dash-text-2xl: 1.35rem;
        --dash-text-3xl: 1.7rem;
        --dash-icon-size: 1.15rem;
    }
    html.dashboard-font-large {
        --dash-text-xs: 0.82rem;
        --dash-text-sm: 1.1rem;
        --dash-text-md: 1.2rem;
        --dash-text-lg: 1.34rem;
        --dash-text-xl: 1.4rem;
        --dash-text-2xl: 1.75rem;
        --dash-text-3xl: 2.1rem;
        --dash-icon-size: 1.45rem;
```

Appendix 1. continued

```
    }
    /* ---------- 3. Font style choices ---------- */
    html.dashboard-font-system {
        --dash-font-family: "Poppins", -apple-system, BlinkMacSystemFont,
                            "Segoe UI", Roboto, sans-serif;
    }
    html.dashboard-font-serif {
        --dash-font-family: "Georgia", "Times New Roman", Times, serif;
    }
    html.dashboard-font-sans-serif {
```

Appendix 1. Continued

```
  --dash-font-family: Arial, Helvetica, sans-serif;
    }
    html.dashboard-font-monospace {
      --dash-font-family: "SFMono-Regular", Consolas, "Liberation Mono",
                          Menlo, monospace;
    }
    /* ---------- 4. Apply the chosen text size to the dashboard ---------- */
    #dashboard-shell .text-xs {
      font-size: var(--dash-text-xs) !important;
      line-height: 1.35rem !important;
    }
    #dashboard-shell .text-sm {
      font-size: var(--dash-text-sm) !important;
      line-height: 1.45rem !important;
     }
    #dashboard-shell .text-base {
      font-size: var(--dash-text-md) !important;
      line-height: 1.52rem !important;
    }
    /* .text-lg, .text-xl, .text-2xl and .text-3xl follow the same pattern */
    #dashboard-shell h1, #dashboard-shell h2, #dashboard-shell h3,
    #dashboard-shell h4, #dashboard-shell h5, #dashboard-shell h6 {
      font-size: var(--dash-text-lg) !important;
    }
    #dashboard-shell p, #dashboard-shell span, #dashboard-shell a,
    #dashboard-shell li, #dashboard-shell td, #dashboard-shell th,
    #dashboard-shell button, #dashboard-shell input,
```

Appendix 1. Continued

```
  #dashboard-shell select, #dashboard-shell textarea {
      font-size: var(--dash-text-sm) !important;
    }
    /* Sidebar and header text and icons */
    #dashboard-shell .sidebar-link {
      font-size: var(--dash-text-sm) !important;
    }
    #dashboard-shell .sidebar-icon-wrap svg,
    #dashboard-shell #app-header button svg {
      width: var(--dash-icon-size) !important;
    height: var(--dash-icon-size) !important;
    }
    #dashboard-shell #app-header h1 {
      font-size: var(--dash-text-lg) !important;
      line-height: 1.35 !important;
    }
    /* ---------- 5. "Reduce animations" setting ---------- */
    html.no-animations *,
    html.no-animations *::before,
    html.no-animations *::after {
      transition: none !important;
      animation: none !important;
      animation-duration: 0s !important;
    }
    /* ---------- 6. Dark mode ---------- */
    /* Colors, shared with the Beneficiary and Volunteer dashboards */
    html.dark-mode #dashboard-shell {
```

Appendix 1. Continued

```
        --dm-bg-base:        #0D0D0D;
        --dm-bg-primary:     #141414;
        --dm-bg-card:        #1E1E1E;
        --dm-bg-elevated:    #252525;
        --dm-bg-subtle:      #2C2C2C;
        --dm-text-primary:   #F0F0F0;
        --dm-text-secondary: #9A9A9A;
        --dm-text-muted:     #5C5C5C;
        --dm-border:         rgba(255, 255, 255, 0.07);
        --dm-border-strong:  rgba(255, 255, 255, 0.12);
        --dm-accent:         #1D9E75;
        --dm-accent-text:    #2ECFA0;
        --dm-accent-subtle:  #0F2E22;
    }
    /* Page background and text */
    html.dark-mode,
    html.dark-mode body,
    body.dark-mode {
        background-color: #0D0D0D !important;
        color: #F0F0F0 !important;
        color-scheme: dark;
        transition: background-color 0.25s ease, color 0.2s ease,
                    border-color 0.2s ease;
    }
    html.dark-mode #dashboard-shell,
    body.dark-mode #dashboard-shell,
    html.dark-mode #dashboard-main-wrapper,
```

Appendix 1. Continued

```
    body.dark-mode #dashboard-main-wrapper,
    html.dark-mode #dashboard-content-scroll,
    body.dark-mode #dashboard-content-scroll {
    background: var(--dm-bg-primary) !important;
    color: var(--dm-text-primary) !important;
}
/* Sidebar and header */
html.dark-mode #dashboard-shell #dashboard-sidebar,
body.dark-mode #dashboard-shell #dashboard-sidebar {
    background: #111111 !important;
    border-color: rgba(255, 255, 255, 0.07) !important;
}
html.dark-mode #dashboard-shell #app-header,
body.dark-mode #dashboard-shell #app-header {
    background: #111111 !important;
    border-bottom-color: rgba(255, 255, 255, 0.06) !important;
}
/* Search box */
html.dark-mode #dashboard-shell #dashboard-search,
body.dark-mode #dashboard-shell #dashboard-search {
    background: #1E1E1E !important;
    border-color: rgba(255, 255, 255, 0.08) !important;
    color: var(--dm-text-primary) !important;
}
/* ... remaining dark mode rules (smooth transitions) continue ... */
```

Appendix 2. Evaluation Instrument

EVALUATION INSTRUMENT FOR: DEVELOPMENT OF WEB-BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT

Name (Optional):_____Date: _____

Address (Optional): _____

Profession: _____Specialization:_____

_____IT expert _____Parents

_____Admins _____ Others

_____Interns

Instructions: Please evaluate using the given scale and placing a check (✓) on the appropriate column corresponding to your response.

Numerical Rating: 5 – excellent **4** – Very Good **3** – Good **2** – Fair **1** – Poor

INDICATORS	5	4	3	2	1
Functionality					
1. Functional completeness – Degree to which the set of functions covers all the specified task and user objectives.					
2. Functional correctness – Degree to which a product or system providles the correct results with the needed degree of precision.					
3. Functional appropriateness – Degree to which the functions facilitate the accomplishment of specified task and objectives.					
A. Performance Efficiency					
1. Time behaviour – Degree to which the response and processing times and throughput rates of a product or system, when performing itis function, meet requirements.					
2. Resource utilization – Degree to which the amounts and types of resources used by a product or system, when performing ills functions, meet requirements.					

Appendix 2. Continued

3. Capacity – Degree to which the maximum limits of a product or system parameter meet requirements.					
B. Compatibility					

1. Co-existence – Degree to which a product can perform it's required functions efficiently while sharing a common environment and resources with other products, without detrimental impact on any other product.					
2. Interoperability – Degree to which two or more systems, products or components can exchange information and use the information that has been exchanged.					

C. Usability					
1. Appropriateness recognisability – Degree to which users can recognize whether a product or system is appropriate for their needs.					
2. Learnability – Degree to which a product or system can be used by specified users to achieve specified goals or learning to use product or system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
3. Operability – Degree to which a product or system has attributes that make it easy to operate and control.					
4. User error protection – Degree to which a system protects users against making errors					
5. User interface aesthetics – Degree to which a user interface enables pleasing and satisfying interaction for user.					

Appendix 2. Continued

6. Accessibility – Degree to which a product or system can be used by people with the widest range of characteristics and capabilities to achieve a specified goal in a specified context of use.					
D. Reliability					
1. Maturity – Degree to which a system, product or component meets needs for reliability under normal operation.					
2. Availability – Degree to which a system, product or component is operational and accessible when required for use.					
3. Fault tolerance – Degree to which a system, product or component operates as intended despite the presence of hardware or software faults.					
4. Recoverability – Degree to which, in the event of an interruption or a failure, product or system can recover the data directly affected and re-establish the desired state of the system.					
E. Security					
1. Confidentiality – Degree to which a product or system ensures that the data are accessible only to those authorized to have access.					
2. Integrity – Degree to which a system, product or component prevents unauthorized access to, or modification of, computer programs or data.					
3. Non-repudiation – Degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later.					
4. Accountability – Degree to which the actions of an entity can be traced uniquely to the entity.					
5. Authenticity – Degree to which the identity of a subject or resource can be proven to be the one claimed.					

Appendix 2. Continued

F. Maintainability					
1. Modularity – Degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components.					
2. Reusability – Degree to which an asset can be used in more than one system, or in building other assets.					
3. Analysability – Degree of effectiveness and efficiency with which is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or cause of failures, or to identity parts to be modified.					
4. Modifiability – Degree to which a product or system can be effectively and efficiently modified without introducing defects or disregarding existing product quality.					
5. Testability – Degree of effectiveness and efficiency with which test criteria can be established for a system, product or component and test can be performed to determine whether those criteria have been met.					
G. Portability					
1. Adaptability – Degree to which a product or system can effectively and efficiently be adapted Fr different or evolving hardware, software or other operational or usage environments.					
2. Installability – Degree of effectiveness and efficiency with which a product or system can be successfully installed and/or uninstalled in a specified environment.					
3. Replaceability – Degree to which a product can replace another specified software product for the same purpose in the same environment.					

**Adopted from ISO 25010*

Remarks/Comments/Suggestions:

_____

_____

_____

Name and Signature of Respondents

Appendix 3. Sample input/output/reports

Appendix Table 1. User Account Management

INPUT	OUTPUT	REMARKS
1. User registration	The system records the user's information and creates an account.	allows new users to access the system.
2. User login	The system verifies the login credentials and opens the page that matches the user's role.	Ensures that only registered users can access their accounts.
3. Updating profile information	The system saves and displays the updated profile.	Ensures that the user's information is accurate and updated.
4. User logout	The system ends the session and returns to the login page.	Protects the account when the user is done using the system.

Appendix Table 2. Volunteer Management

INPUT	OUT	REMARKS
1. Submitting a volunteer application	The system records the application and shows it as pending.	Allows interested users to apply as volunteers.
2. Approving a Volunteer	The system updates the status of the volunteer and gives access to volunteer features.	Ensures that only approved volunteers can join activities.
3. Assigning a volunteer to a task	The system saves the assignment and displays it in the volunteer's account.	Helps the administrators organize the volunteers.
4. Viewing assigned activities	The system displays the activities assigned to the volunteer.	Allows volunteers to know their schedule and tasks.

Appendix 2. Continue

5. Viewing participation history	The system displays the activities previously joined by the volunteer.	Allows volunteers to keep track of their participation.

Appendix Table 3. Activity Management

INPUT	OUTPUT	REMARKS
1. Creating an activity	The system saves the activity with its title, date, and location.	Allows the administrators to plan the projects of the organization.
2. Updating the activity schedule	The system saves and displays the updated schedule.	Keeps the schedule accurate for volunteers and beneficiaries.
3. Updating the activity Status	The system displays the new status of the selected activity.	Helps in monitoring ongoing and completed activities.
4. Recording participation	The system records the attendance of the volunteers in the activity.	Provides the basis for the participation records and certificates.
5. Viewing the list of activities	The system displays the upcoming and past activities.	Allows users to know the activities of the organization.

Appendix Table 4. Donation Management

INPUT	OUTPUT	REMARKS
1. Recording a donation Received	The system saves the donor, the type, and the amount or quantity of the donation.	Keeps an accurate record of the donations.

Appendix 3. Continue

2. Recording a donation given out	The system saves the recipient and the items distributed, and updates the inventory.	Shows where the donations were used.
3. Viewing the donation records	The system displays the list of donations received and given.	Allows the administrators to monitor the donations.
4. Viewing the inventory	The system displays the remaining items in stock.	Helps in planning the distribution of goods.

Appendix Table 5. Beneficiary Management

INPUT	OUTPUT	REMARKS
1. Registering a beneficiary	The system saves the beneficiary's profile.	Allows the organization to keep records of the people it serves.
2. Verifying a beneficiary	The system updates the status of the selected beneficiary.	Ensures that the records are checked by the administrators.
3. Updating beneficiary information	The system saves and displays the updated profile.	Ensures that the information of the beneficiary is accurate.
4. Viewing joined programs	The system displays the programs and activities joined by the beneficiary.	Allows beneficiaries to see the help they received.

Appendix Table 6. Reports and Dashboard

INPUT	OUTPUT	REMARKS

1. Viewing the Dashboard	The system displays the latest data on donations,volunteers, and activities.	Gives the administrators a quick view of the status of the project.
2. Generating a report	The system compiles the selected records into a report.	Reduces the time spent in preparing reports by hand.
3. Printing or saving a report as PDF	The system produces a printable or downloadable copy of the report.	Allows the report to be shared and kept.
4. Generating a participation certificate	The system creates a certificate with the volunteer's name and activity.	Recognizes the participation of the volunteers.

Appendix Table 7. E-Commerce

INPUT	OUTPUT	REMARKS
1. Viewing the products	The system displays the available products and their prices.	Allows customers to see what the community sells.
2. Placing an order	The system records the order and its details.	Supports the sales that fund the project.
3. Adding or updating a Product	The system saves and displays the product information.	Allows the administrators to keep the store updated.
4. Updating the order status	The system displays the new status of the selected order.	Helps in monitoring the orders until they are completed.

Appendix Table 8. Community Engagement

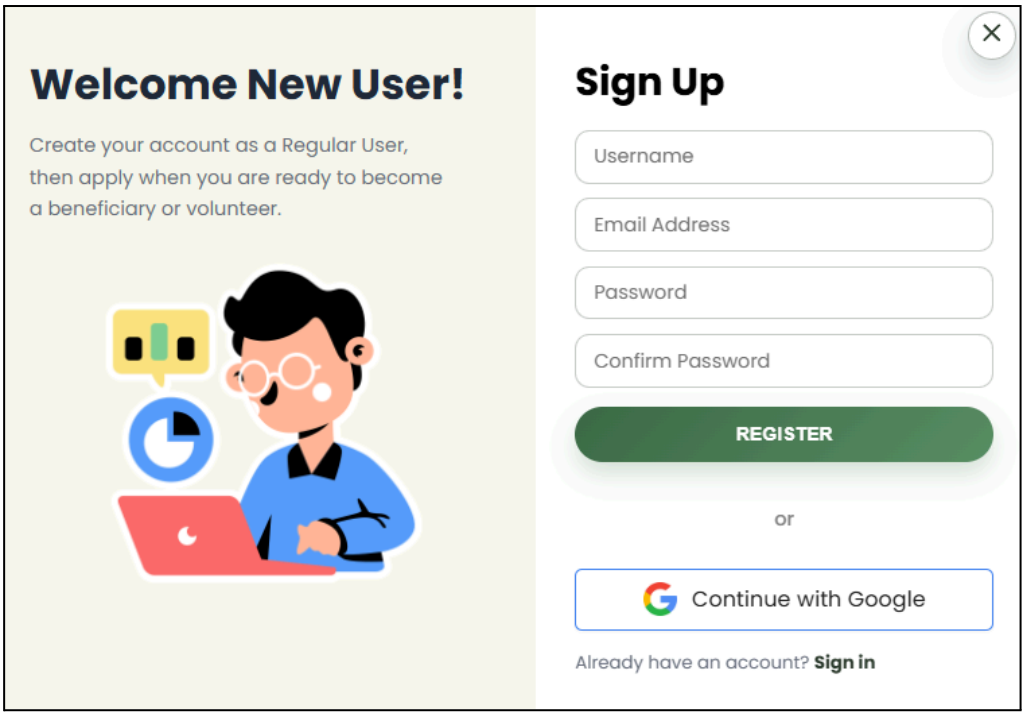
INPUT	OUTPUT	REMARKS

1. Viewing announcements	The system displays the latest announcements of the organization.	Keeps the users informed about the programs.
2. Submitting feedback	The system records the feedback of the user.	Allows users to share their comments and suggestions.
3. Reviewing feedback	The system displays the submitted feedback to the administrators.	Allows the administrators to identify user concerns.

Appendix 4. Users' Manual Guide

1. Click Sign In, then click Sign up.
2. Type your Username, Email Address, Password, and Confirm Password.
3. Click REGISTER, or click Continue with Google.

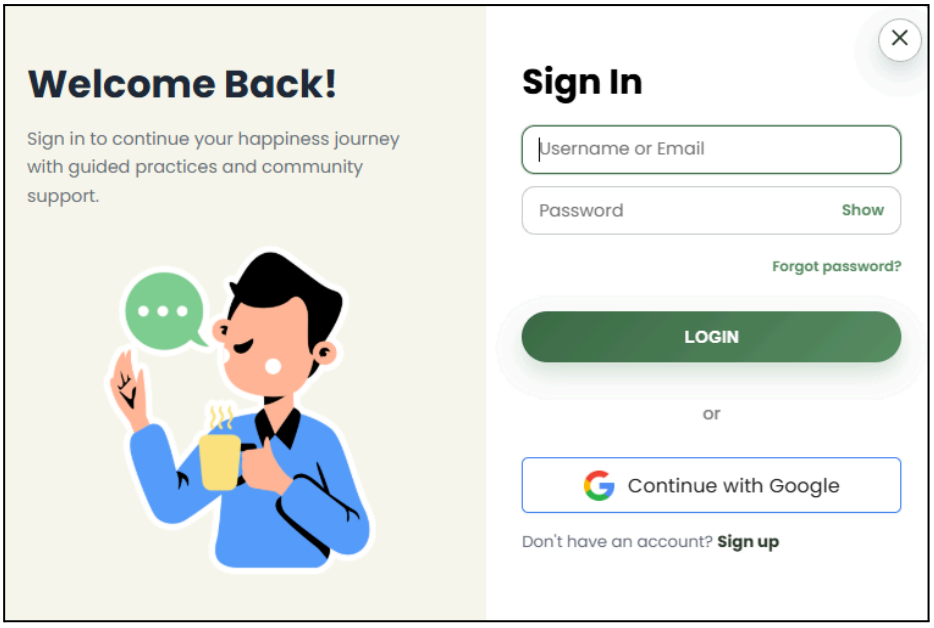
   Every new account starts as a Regular User.



Appendix Figure 1. Sign Up Page

Sign in

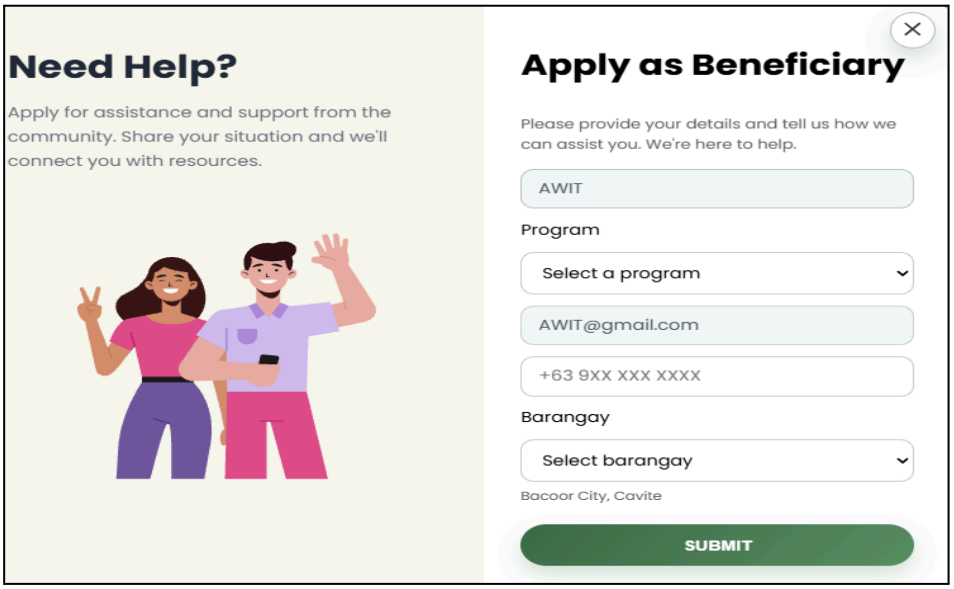
1. Click Sign In.
2. Type your Username or Email and your Password.
3. Click LOGIN, or click Continue with Google.



Appendix Figure 2. Sign In Page

Apply as a beneficiary

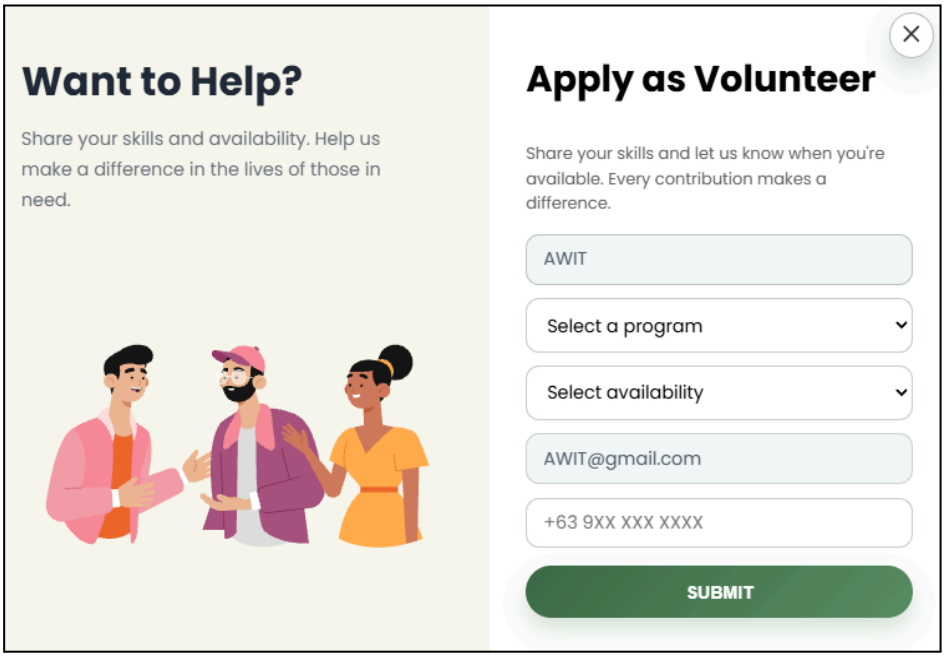
1. On the landing page, scroll down and click Apply as Beneficiary.
2. Type your Full name, Email, and Phone number.
3. Choose your Program and your Barangay.
4. Click SUBMIT.



Appendix Figure 3. Apply as Beneficiary Form

Apply as a volunteer

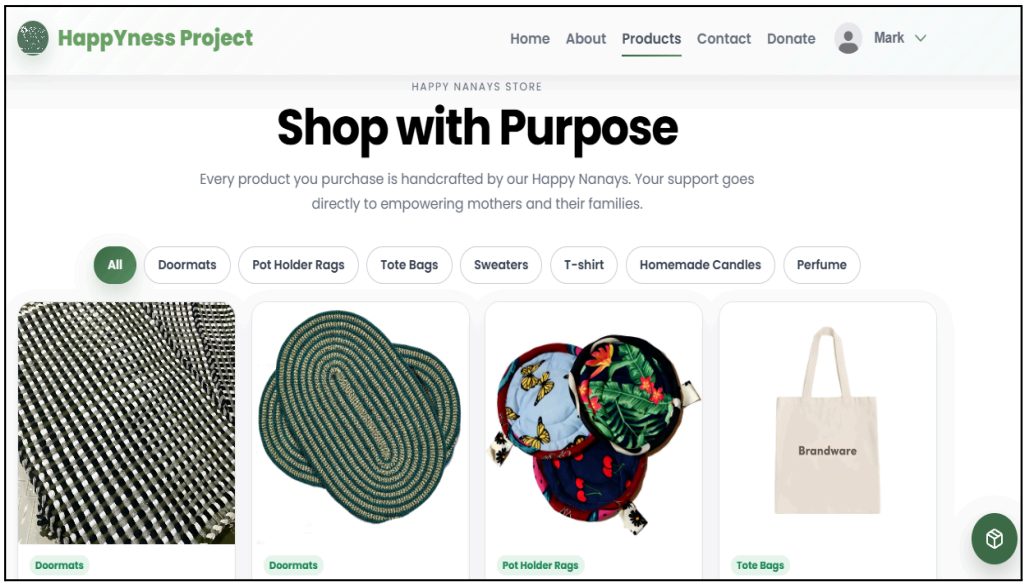
1. On the landing page, scroll down and click Apply as Volunteer.
2. Type your Full name, Email, and Phone number.
3. Choose your Program (Intern or Community Volunteer) and your Availability.
4. Click SUBMIT.



Appendix Figure 4. Apply as Volunteer Form

Buy a product

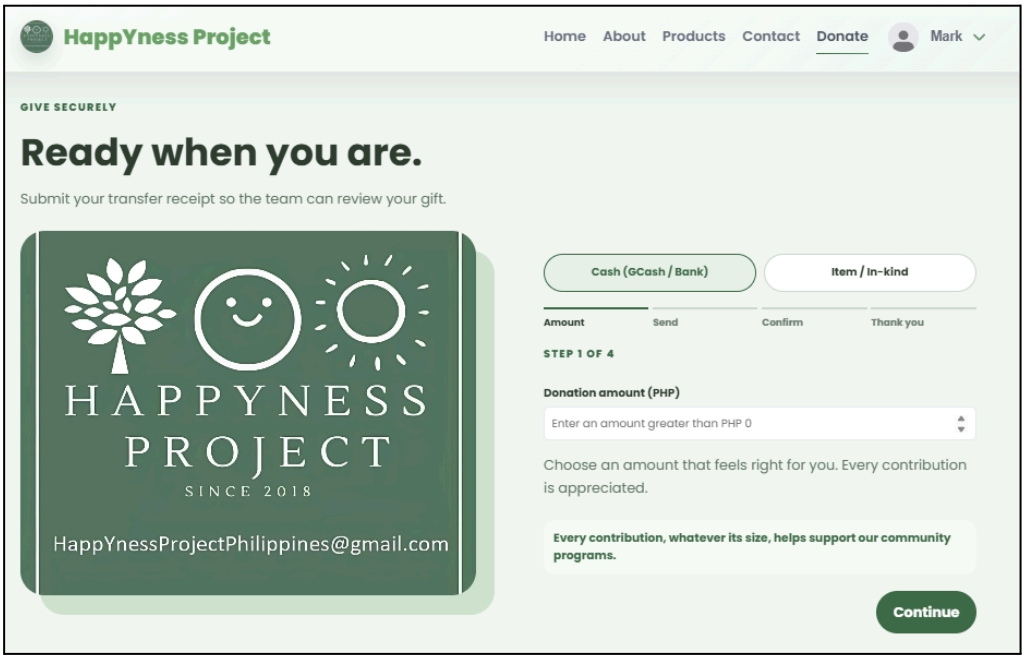
1. Click Products and choose an item.
2. Set the quantity and click Continue.
3. Choose a Payment method: COD, GCash, or Bank Transfer.
4. For GCash or Bank Transfer, send the payment and upload your proof of payment.
5. Click Place order.



Appendix Figure 5. Products Page

Make a donation

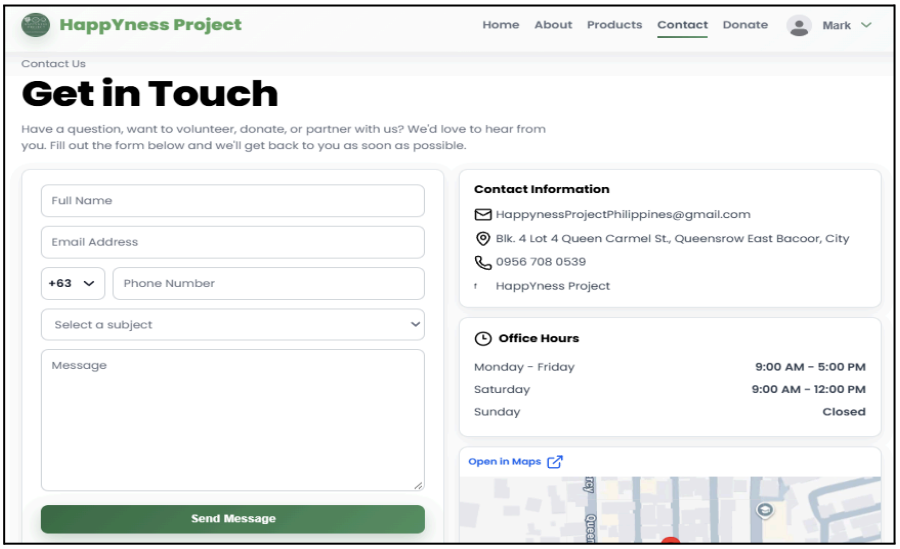
1. Click Donate.
2. Type the Donation amount and click Continue.
3. Send your donation and upload your receipt
4. Wait for the Thank you page. The team will check your receipt.



Appendix Figure 6. Donate Page

Send a message

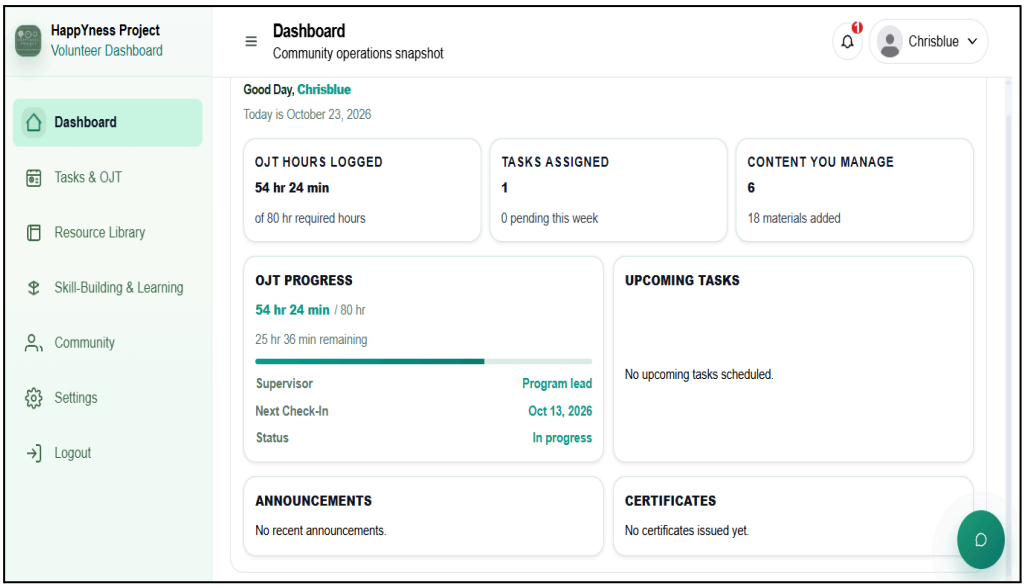
1. Click Contact.
2. Fill in your name, email, phone number, subject, and message.
3. Click Send Message.



Appendix Figure 7. Contact Page

Volunteer

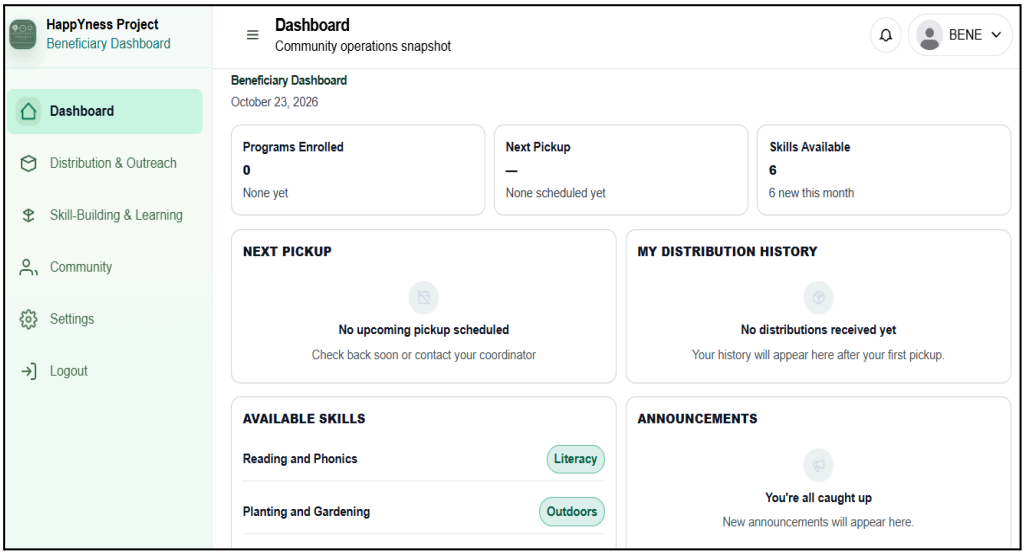
1. Sign in with your volunteer account.
2. Open your assigned activities to see your tasks and schedule.
3. Record your participation after each activity.
4. Open your participation history to view or download your certificate.



Appendix Figure 8. Volunteer Pages

Beneficiary

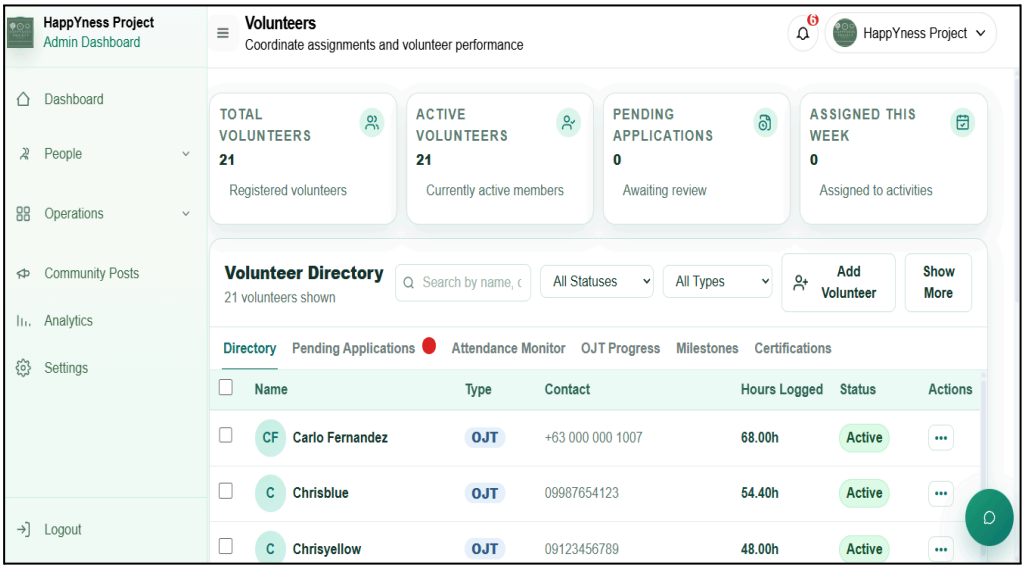
1. Sign in with your beneficiary account.
2. Update your profile if your information changed.
3. Open your records to see the help you received.



Appendix Figure 9. Beneficiary Pages

Admin

1. Sign in with the admin account. The Dashboard shows the latest data.
2. Volunteers: approve applicants and assign tasks.
3. Activities: add activities and update their schedule and status.
4. Donations: record donations received and given out.
5. Beneficiaries: register and verify beneficiaries.
6. Products and Orders: add products and update order status.
7. Reports: choose a report and date range, then print or save it as PDF.



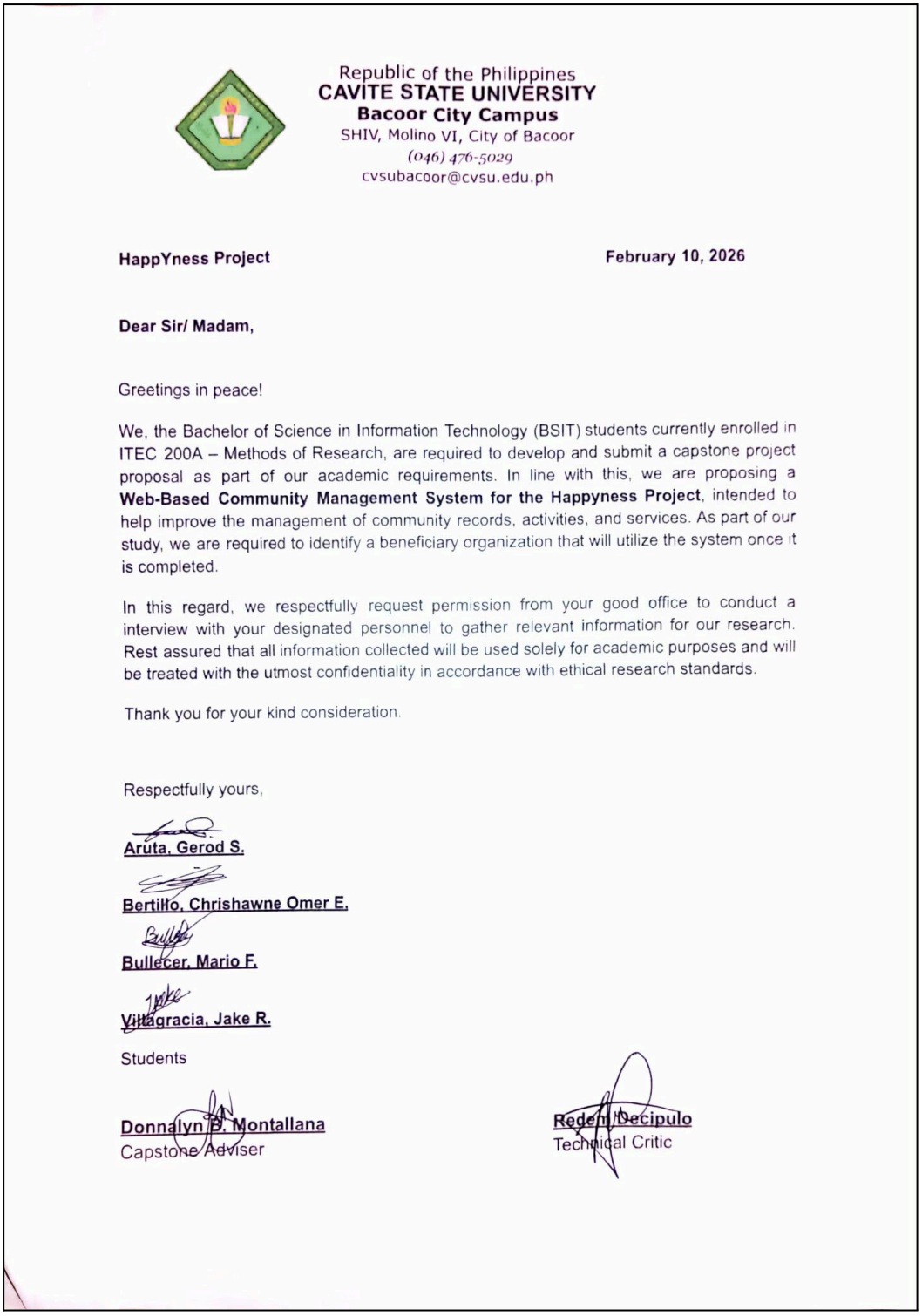
Appendix Figure 10. Admin Dashboard

Appendix 5. Gantt Chart Template

ACTIVITY	2025			2026										
	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Proposal drafting														
Proposal defense														
System design														
System development														
System testing														
System evaluation														
System implementation														
Manuscript preparation														
Final oral defense														
Manuscript review														

Appendix Figure 1. Gantt Chart of Web-Based Community Management System

Appendix 6. Proposal Letter for Client



Republic of the Philippines
CAVITE STATE UNIVERSITY
Bacoor City Campus
SHIV, Molino VI, City of Bacoor
(046) 476-5029
cvsubacoor@cvsu.edu.ph

HappYness Project **February 10, 2026**

Dear Sir/ Madam,

Greetings in peace!

We, the Bachelor of Science in Information Technology (BSIT) students currently enrolled in ITEC 200A – Methods of Research, are required to develop and submit a capstone project proposal as part of our academic requirements. In line with this, we are proposing a **Web-Based Community Management System for the Happyness Project**, intended to help improve the management of community records, activities, and services. As part of our study, we are required to identify a beneficiary organization that will utilize the system once it is completed.

In this regard, we respectfully request permission from your good office to conduct a interview with your designated personnel to gather relevant information for our research. Rest assured that all information collected will be used solely for academic purposes and will be treated with the utmost confidentiality in accordance with ethical research standards.

Thank you for your kind consideration.

Respectfully yours,

Aruta, Gerod S.

Bertillo, Chrishawne Omer E.

Bullecer, Mario F.

Villagracia, Jake R.

Students

Donnalyn B. Montallana
Capstone Adviser

Redem Decipulo
Technical Critic

Appendix Figure 2. Proposal Letter for Client

Appendix 7. Interview Report

RAW INTERVIEW

Interviewer: Good morning! We are here to develop a system for you as part of our thesis project.

Interviewee: Good morning as well. That sounds great; it would be a huge help to us.

Interviewer: What features would you like to include in the system? What are your specific requirements?

Interviewee: Ideally, we want the system to handle donations, teaching programs for children, planting initiatives, rag-making, volunteer tracking, and e-commerce.

Interviewer: Noted. Regarding the rag-making and planting, will these products be posted on the system for sale?

Interviewee: Yes, that's correct. That is what the "Happy-Nanays" (Happy Moms) and the children currently produce.

Interviewer: So, we will create a dedicated product page where all the rags and plants made by the children will be displayed.

Interviewer: For the volunteers, they will have separate accounts, while the "full access" or administrative control will belong to you as the Admin.

Interviewee: I see. So they will just register, but will they have access to their task assignments?

Interviewer: Yes. They can update their profiles and access assigned activities and data. However, as the Admin, you will have total access to everything, including the volunteer records.

Interviewee: Is it possible for us to have a dashboard?

Interviewer: Definitely. We will include a real-time dashboard that displays live data on donations, volunteer participation, and the status of various programs.

Interviewee: That sounds good. How about the activities?

Appendix Figure 3. Page 1 of Interview Report

Interviewer: Yes, there will also be a section for ongoing activities and beneficiary tracking.

Interviewer: We will also include an activity tracking module where you can monitor ongoing programs, especially those focused on education.

Interviewee: And will I be able to access all of that as the Admin?

Interviewer: Yes, you will have full administrative access. You will also have the authority to create programs and assign them to specific volunteers.

Interviewee: That would be excellent.

Interviewer: Furthermore, regarding the volunteers, your approval as the Admin will be required for their registration and for their participation in specific activities



Signature: Justine Joy Paraiso, RSW

Appendix Figure 4. Page 2 of Interview Report

INTERVIEW REPORT

Item	Details
Client	HappYness Project
Interviewee(s)	Admin/Staff & Volunteers
Interviewer(s)	Aruta, Bertillo, Bullecer, Villagracia
Date	October 10, 2026
Mode	Face-to-face

1. Purpose of the Interview

The interview was held to gather the client's feedback and requirements for improving the HappYness Project system used by admins, volunteers, beneficiaries, and donors. The discussion covered donations and expenses, skill building, account management, admin tools, website design, and feedback.

2. Detailed Discussion

2.1 Funds and Donations

Topic 1: Use of donations and event funding

The interviewer wanted to clarify where money from donations is used. According to one of the admins, regular activities such as tutorials are mostly funded by their boss, Wewe, although donations from other people sometimes cover items like snacks. Big events are usually funded by donations from partner organizations, companies, and individuals. This gives a clear picture of how donations are spent.

Topic 2: Licensing, partners, and sponsorships on the Donate page

The Donate page does not yet show the organization's credentials and supporters, so the client asked to display licensing, partners, and sponsorships on it. This will make the page more credible and better presented.

Topic 3: Item or product donations

Some donors give products or items instead of money. The client agreed to enhance the Donate page to handle item donations, so donors can give in-kind donations through the system.

2.2 Beneficiaries and Volunteers

Topic 4: Volunteer record of skill-building progress

Appendix Figure 5. Page 3 of Interview Report

Beneficiaries' progress on the learning materials needs to be managed. Each volunteer will have an individual record to track the progress of their beneficiaries, which gives clear tracking of skill-building progress.

Topic 5: Admin approval for personal information edits

Beneficiaries should not change their personal information freely. Admin approval is required before their edits take effect, which keeps records accurate and verified.

Topic 6: Account classifications

Accounts are not yet grouped by user type. The client agreed to classify accounts based on beneficiary and volunteer types, which makes accounts easier to manage and organize.

Topic 7: Common feedback questions after activities

Beneficiary feedback needs more structure. The client asked to add general or common questions for the audience to answer after an activity or event, so feedback is more consistent and useful.

2.3 Admin Tools

Topic 8: View options for applications

The admin needs an easier way to review applications. View options will be added for beneficiary and volunteer applications on the admin side, for faster and clearer review.

Topic 9: Admin-editable landing page

The landing page currently needs developer help to change. The admin should be able to edit its layout and design without the developers, so updates can be made independently.

Topic 10: Record of admin actions

Actions made by other admins on operations are not traceable. A record of these actions will be added for better accountability.

2.4 Website Design and Access

Topic 11: Images in the "Why Choose HappYness Project" section

The section uses icons, which look plain. The client asked to replace them with images to make the section more visually appealing.

Topic 12: Forgot email password

Users who forget their email password need a way to regain access. A recovery option will be added so they can do so without admin help.

3. Conclusion

The client's feedback focused on improving the Donate page, tracking beneficiary skill building, tightening control over account information, giving the admin more tools and

Appendix Figure 6. Page 4 of Interview Report

accountability, and making the website more visually appealing. These requirements will guide the next phase of development.



Prepared by:
Aruta, Gerod S.
Bertillo, Chrishawne Omer E.
Bullecer, Mario F.
Villagracia, Jake R.

Noted by:

Justine Joy Paraiso, RSW

Signature of client representative

Appendix Figure 7. Page 5 of Interview Report

INTERVIEW REPORT

Item	Details
Client	HappYness Project
Interviewee(s)	Admin/Staff & Volunteers
Interviewer(s)	Aruta, Bertillo, Bullecer, Villagracia
Date	August 29, 2026
Mode	Online meeting

1. Purpose of the Interview

The interview was held to gather the client's feedback and requirements for improving the outreach management system used by admins, volunteers, beneficiaries, and regular users. The discussion covered account management, scope, activity handling, attendance verification, donations, order tracking, scheduling, feedback, notifications, and reports.

2. Detailed Discussion

2.1 Account Management and Scope

Topic 1: Easier account creation for the admin

Creating accounts for beneficiaries and volunteers takes a lot of the admin's time, especially when many members are registered in one day. The client asked to simplify the process, so the admin can register members with fewer steps and fewer required fields.

Topic 2: System scope limited to Bacoor

Most of the beneficiaries and volunteers are established in Bacoor, where most activities are also held. The client asked to change the scope to Bacoor only, so the system focuses on the area the organization actually serves.

2.2 Activity Management

Topic 3: Selecting specific beneficiaries for an activity

Not every activity is meant for all beneficiaries. The admin will be able to choose specific beneficiaries when setting an activity, so only the intended beneficiaries are invited and notified.

Topic 4: Volunteer verification of beneficiary attendance

There is currently no way to confirm that a beneficiary on the list really attended an activity or event. Assigned volunteers will verify attendance on the day itself, which gives accurate records of who actually received help.

2.3 Donations and Products

Topic 5: Donation page enhancement

Appendix Figure 8. Page 6 of Interview Report

The current donation page needs to look more presentable and be easier to follow. The client agreed to enhance its layout and flow so donors can give with more confidence. The specific details will be discussed during the design stage.

Topic 6: Order tracking modal on the product page

Regular users keep asking where their orders are. A viewing modal will be added on the product page so users can track the progress of their ordered items without asking the admin.

2.4 Scheduling, Feedback, and Notifications

Topic 7: Activity calendar on the volunteer dashboard

Volunteers sometimes forget or get confused about the dates of their activities. An activity calendar will be added to the volunteer dashboard, showing assigned and upcoming activities for better schedule management.

Topic 8: Feedback for previous distribution outreach

There is no channel yet for beneficiaries to share their thoughts after a distribution outreach. The client asked to add feedback for previous outreach, so the admin can view feedback and concerns in the admin dashboard and improve future distributions.

Topic 9: Enhanced email for new activity notifications

The email sent to assigned volunteers and beneficiaries about a new activity looks plain. The client asked to redesign it so key details such as the activity, date, time, and place are clear at a glance.

2.5 Reports

Topic 10: Export and download of certificates and reports

Preparing documents for partners and volunteers takes a lot of time. The admin will be able to export or download certificates and reports generated from the analytics data, which makes documentation faster and more consistent.

3. Suggested Priority

Priority	Items
High	Account creation (1), Bacoor scope (2), Select beneficiaries (3), Attendance verification (4)
Medium	Volunteer calendar (7), Feedback module (8), Reports and certificates (10), Order tracking modal (6)
Low	Donation page enhancement (5), Email design (9)

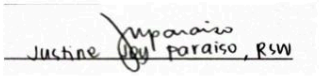
Appendix Figure 9. Page 7 of Interview Report

4. Conclusion

The client's feedback focused on making the system easier for the admin, narrowing its scope to Bacoor, improving the accuracy of activity and attendance records, and strengthening communication, feedback, and reporting. These requirements will guide the next phase of development.

Prepared by:
Aruta, Gerod S.
Bertillo, Chrishawne Omer E.
Bullecer, Mario F.
Villagracia, Jake R.

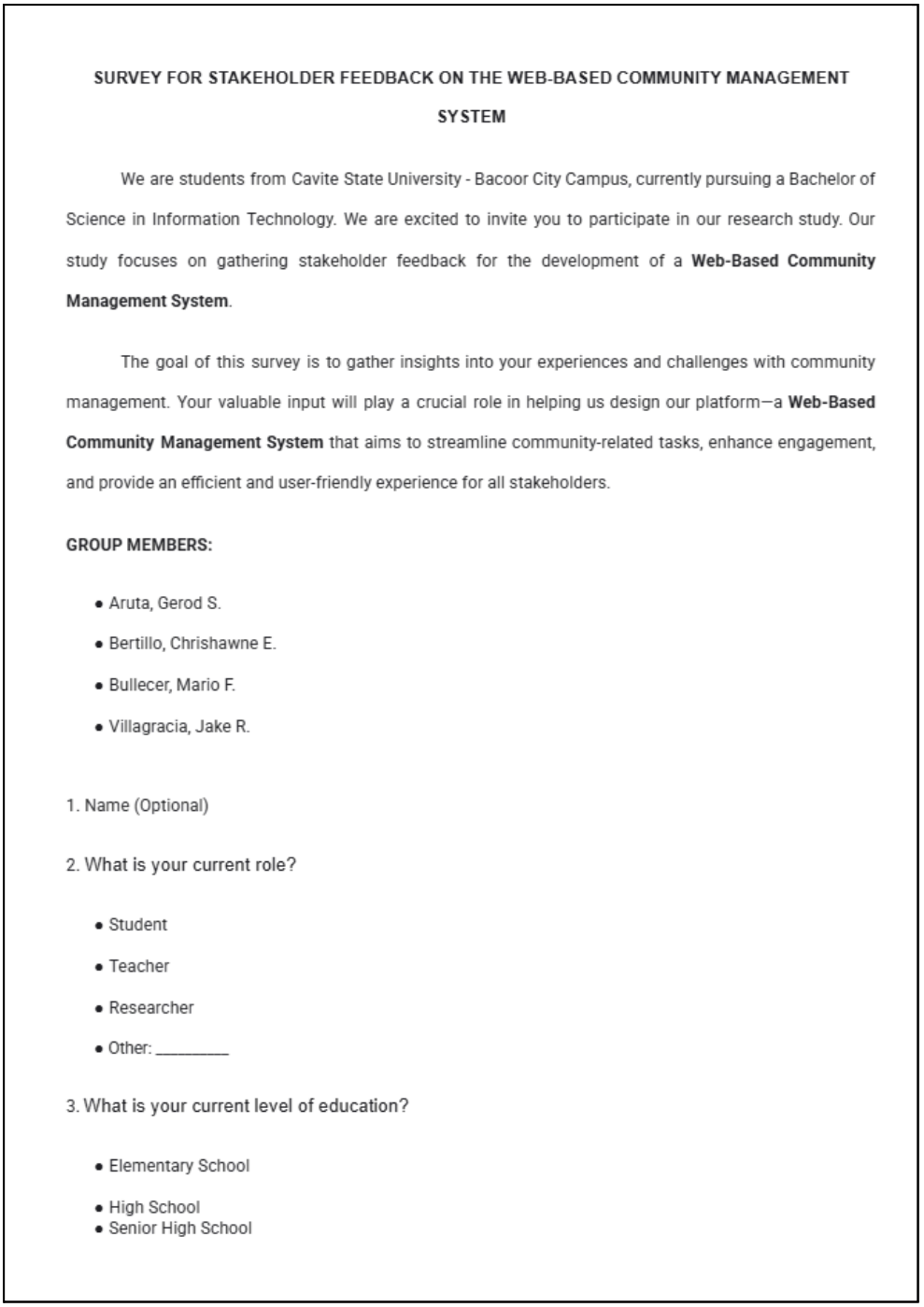
Noted by:



Signature of client representative

Appendix Figure 10. Page 8 of Interview Report

Appendix 8. Survey Questionnaire for the Web-Based Community Management System



SURVEY FOR STAKEHOLDER FEEDBACK ON THE WEB-BASED COMMUNITY MANAGEMENT SYSTEM

We are students from Cavite State University - Bacoor City Campus, currently pursuing a Bachelor of Science in Information Technology. We are excited to invite you to participate in our research study. Our study focuses on gathering stakeholder feedback for the development of a **Web-Based Community Management System**.

The goal of this survey is to gather insights into your experiences and challenges with community management. Your valuable input will play a crucial role in helping us design our platform—a **Web-Based Community Management System** that aims to streamline community-related tasks, enhance engagement, and provide an efficient and user-friendly experience for all stakeholders.

GROUP MEMBERS:

- Aruta, Gerod S.
- Bertillo, Chrishawne E.
- Bullecer, Mario F.
- Villagracia, Jake R.

1. Name (Optional)

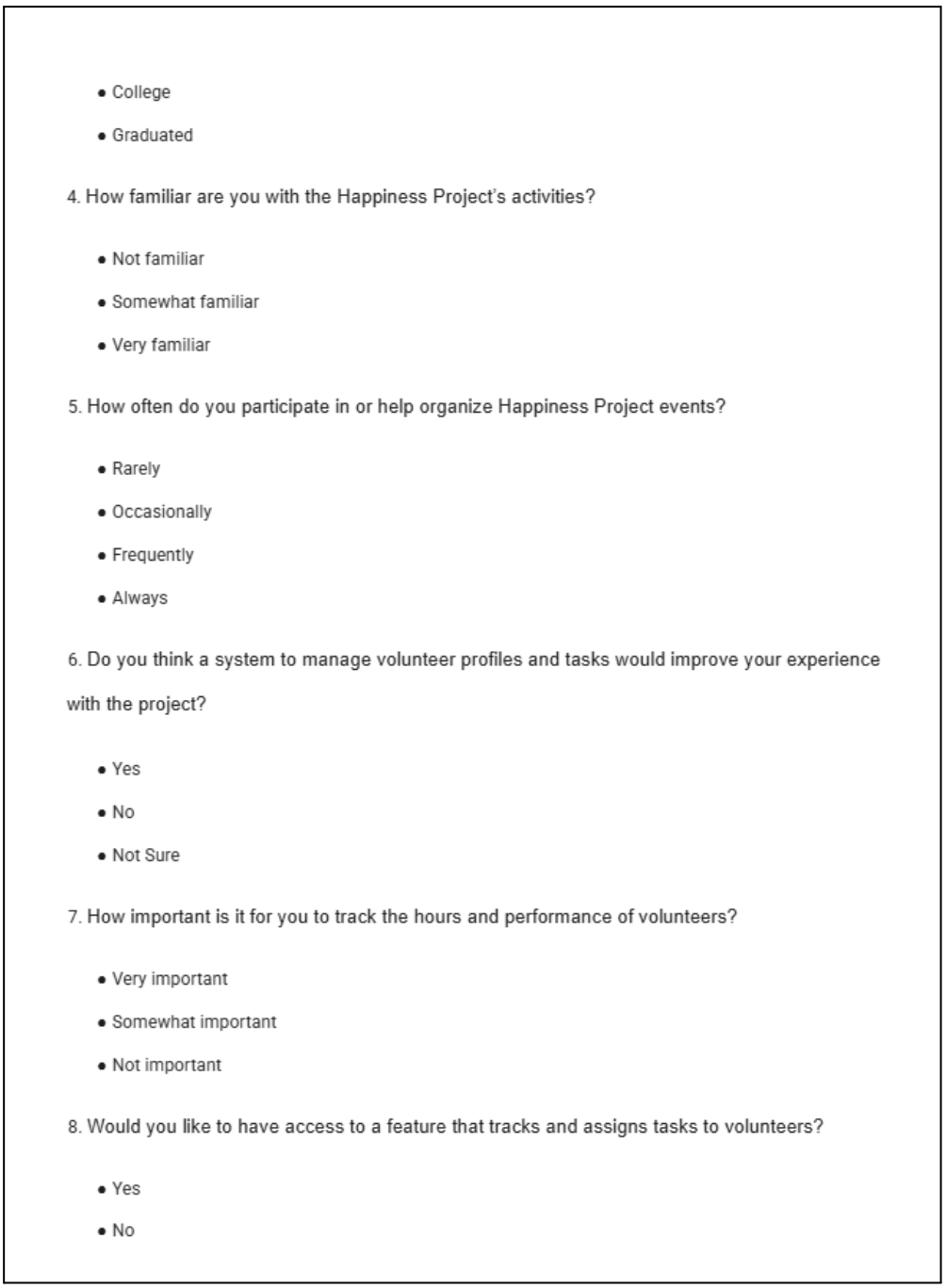
2. What is your current role?

- Student
- Teacher
- Researcher
- Other: _____

3. What is your current level of education?

- Elementary School
- High School
- Senior High School

Appendix Figure 5. Page 1 of Survey Questionnaire for Web-Based Community Management System



- College
- Graduated

4. How familiar are you with the Happiness Project's activities?

- Not familiar
- Somewhat familiar
- Very familiar

5. How often do you participate in or help organize Happiness Project events?

- Rarely
- Occasionally
- Frequently
- Always

6. Do you think a system to manage volunteer profiles and tasks would improve your experience with the project?

- Yes
- No
- Not Sure

7. How important is it for you to track the hours and performance of volunteers?

- Very important
- Somewhat important
- Not important

8. Would you like to have access to a feature that tracks and assigns tasks to volunteers?

- Yes
- No

Appendix Figure 6. Page 2 of Survey Questionnaire for the Web-Based Community Management System

- Not sure

9. Would you use a platform that provides interactive tutorials for rag-making or planting activities?

- Very Likely
- Likely
- Unlikely
- Very Unlikely

10. Would having an interactive calendar for activities and events be useful to you?

- Yes
- No
- Not Sure

11. How helpful would it be to track the progress of rag-making or planting activities through a digital platform?

- Very helpful
- Somewhat helpful
- Not Helpful

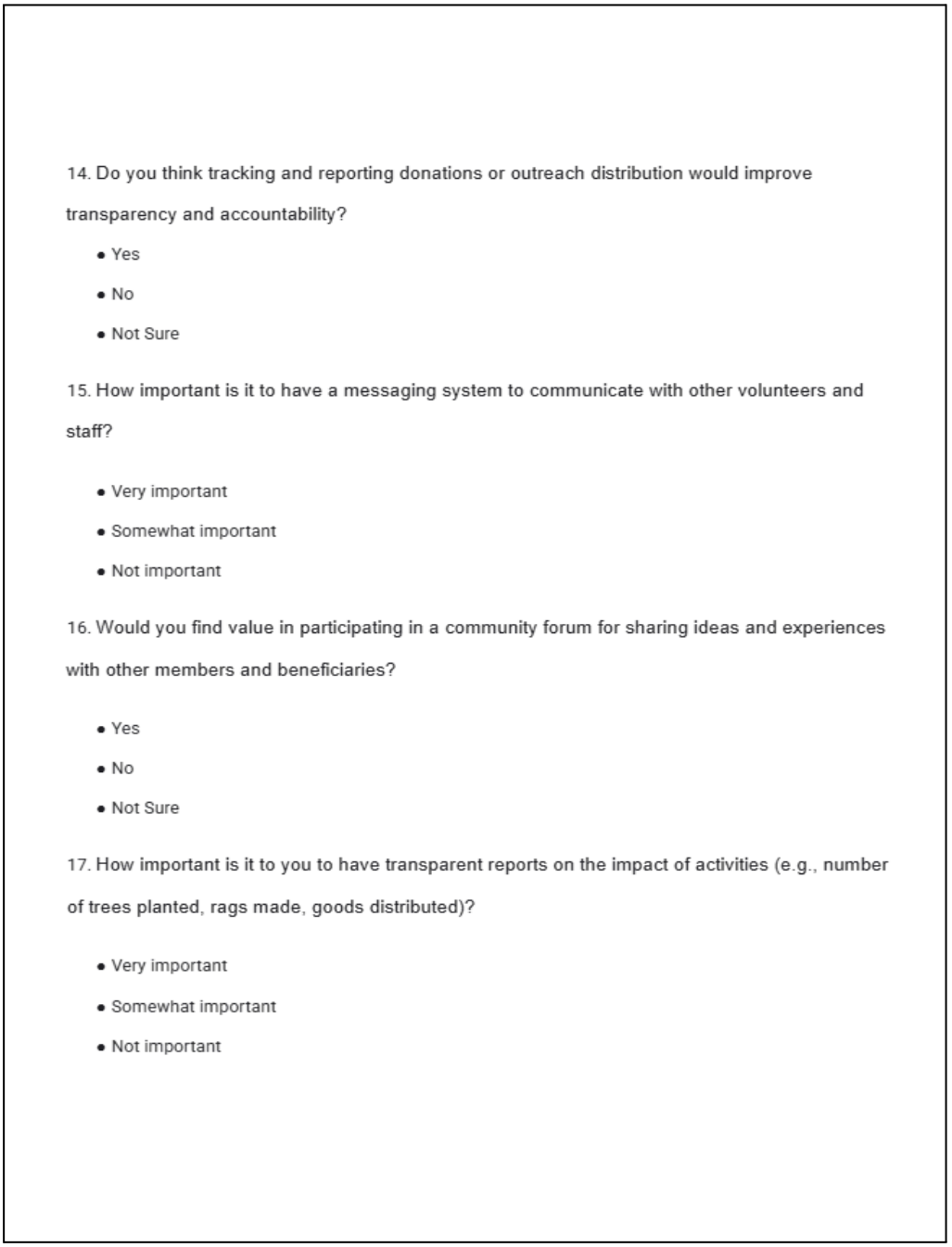
12. How useful would it be to have a feature for scheduling and tracking donation distributions?

- Very useful
- Somewhat useful
- Not useful

13. Would you prefer to receive updates about distribution events via a messaging system?

- Yes
- No
- Not sure

Appendix Figure 7. Page 3 of Survey Questionnaire for the Web-Based Community Management System



14. Do you think tracking and reporting donations or outreach distribution would improve transparency and accountability?

- Yes
- No
- Not Sure

15. How important is it to have a messaging system to communicate with other volunteers and staff?

- Very important
- Somewhat important
- Not important

16. Would you find value in participating in a community forum for sharing ideas and experiences with other members and beneficiaries?

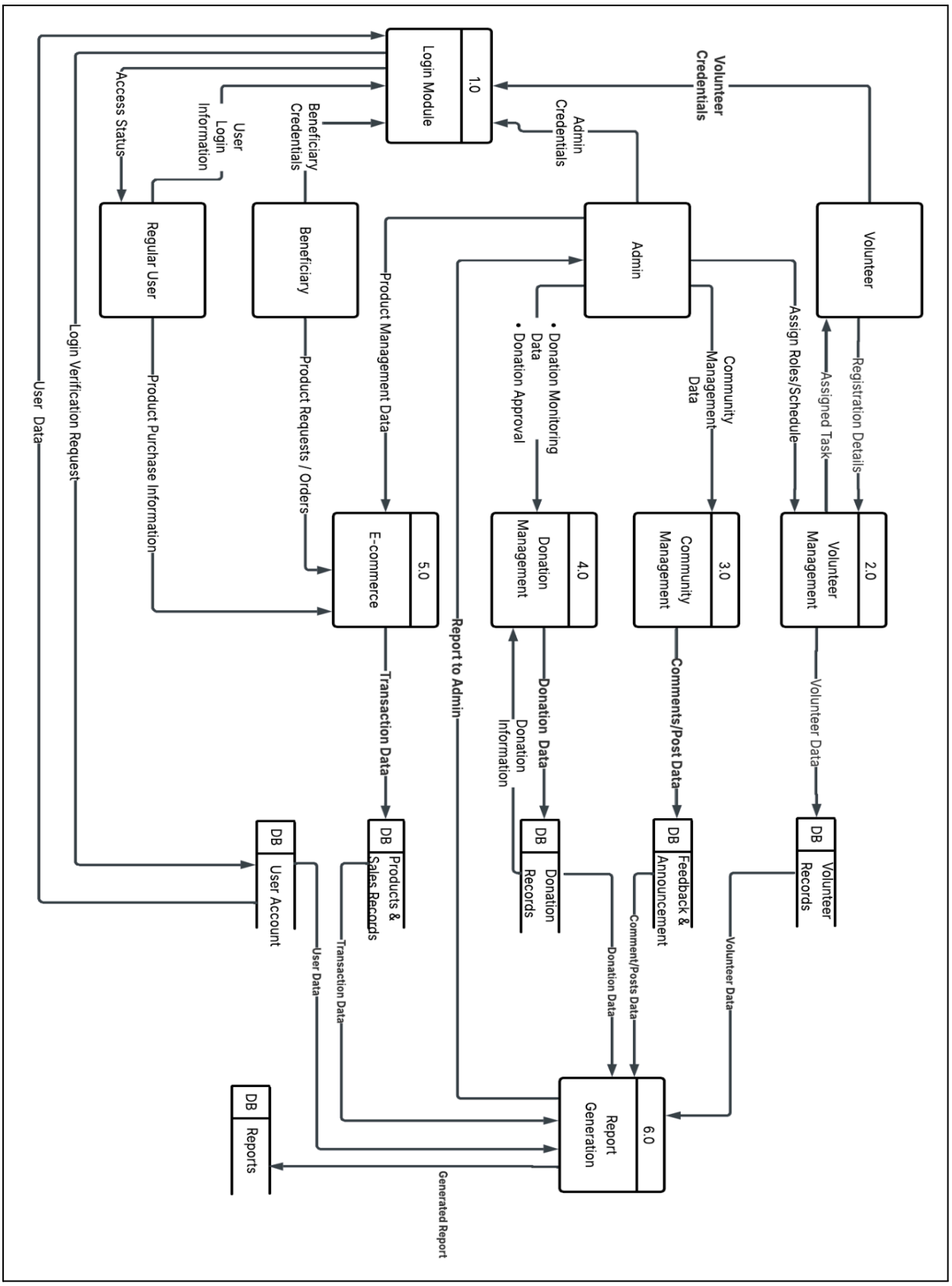
- Yes
- No
- Not Sure

17. How important is it to you to have transparent reports on the impact of activities (e.g., number of trees planted, rags made, goods distributed)?

- Very important
- Somewhat important
- Not important

Appendix Figure 8. Page 4 of Survey Questionnaire for the Web-Based Community Management System

Appendix 9. Data flow Diagram



Appendix 10. Photo Documentation



Photo 1. Presentation of the system and Gathering data from the client

Appendix 10. Continued



Photo 2. Evaluation of IT Expert

Appendix 10. Continued



Photo 3. Evaluation of IT Expert



Photo 4. Evaluation of IT Expert



Photo 5. Identification credentials of the IT experts responsible for evaluating the system

Appendix 10. Continued

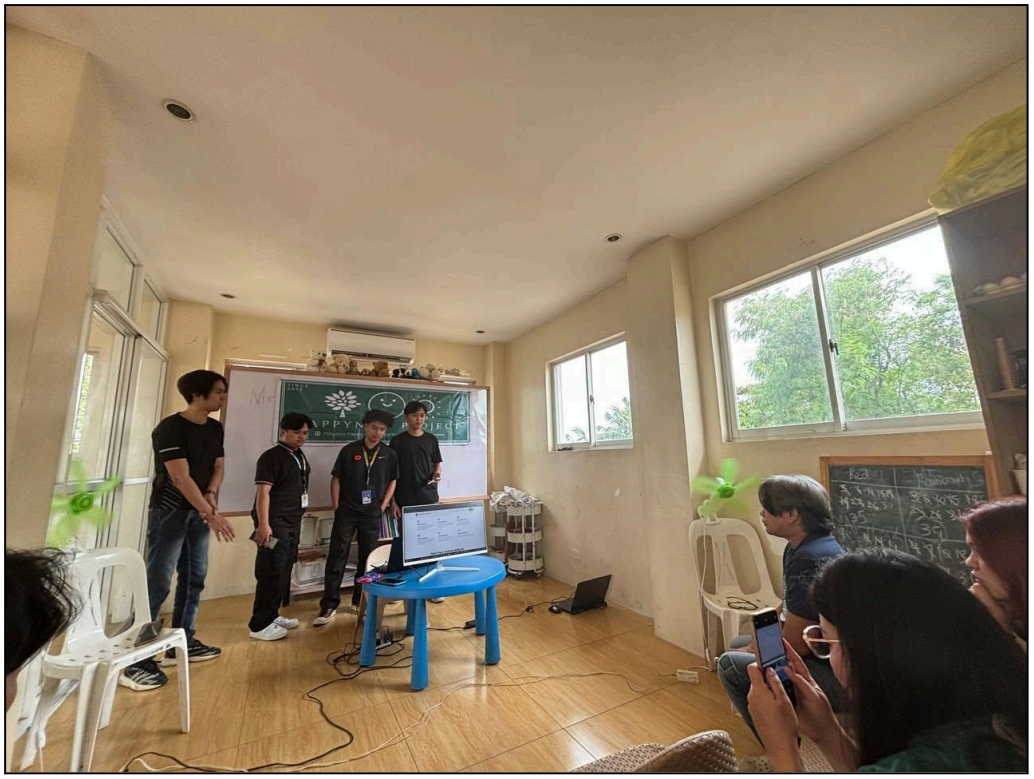


Photo 6. Evaluation of the Organization Admins and Intern



Appendix 10. Continued



Photo 7. Evaluation with the beneficiaries

Appendix 11. Unit Testing

Appendix Table 9. Module: User Account Management

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Regular User Side		
Creating an Account	S	Working
Creating an Account with Google	S	Working
Logging In	S	Working
Logging In with Google	S	Working
Showing and Hiding the Password	S	Working
Resetting the Password	S	Working
Applying as a Beneficiary	S	Working
Applying as a Volunteer	S	Working
Logging Out	S	Working
Volunteer Side		
Logging In	S	Working
Updating Profile Information	S	Working
Logging Out	S	Working
Beneficiary Side		
Logging In	S	Working
Updating Profile Information	S	Working
Logging Out	S	Working
Admin Side		
Logging In	S	Working
Accessing the Admin Dashboard	S	Working
Logging Out	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix Table 10. Module: Volunteer Management

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Regular User / Applicant Side		
Submitting a Volunteer Application	S	Working
Applying as a Volunteer	S	Working
Volunteer Side		
Logging In	S	Working
Viewing Assigned Tasks & Duties	S	Working
Signing Up for Outreach Events	S	Working
Tracking OJT Rendered Hours	S	Working
Downloading Digital Participation Certificates	S	Working
Updating Availability Schedules	S	Working
Logging Out	S	Working
Admin Side		
Reviewing Pending Applications	S	Working
Approving or Rejecting Applicants	S	Working
Assigning Tasks & Outreach Duties	S	Working
Verifying Rendered OJT Hours	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix Table 11. Module: Activity Management

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Admin Side		
Creating a Community Activity	S	Working
Setting Activity Schedule & Location	S	Working
Updating Activity Status	S	Working
Assigning Volunteers to Activities	S	Working
Volunteer Side		
Viewing Event Schedules & Details	S	Working
Recording Attendance & Task Completion	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix Table 12. Module: Donation Management

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Admin Side		
Recording Received Monetary Donations	S	Working
Recording Received In-Kind Goods	S	Working
Logging Item Allocations and Distributions	S	Working
Managing Multi-site Inventory Ledger	S	Working
Generating Donation Receipts	S	Working
Regular User / Donor Side		
Submitting Donation Details & Receipt Proof	S	Working
Viewing Fund Utilization & Reach Reports	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix Table 13. Module: Beneficiary Management

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Regular User / Beneficiary Side		
Submitting Beneficiary Application	S	Working
Applying for Support Programs / Goods	S	Working
Viewing Records of Distributed Goods Received	S	Working
Accessing Educational Tutorials	S	Working
Admin Side		
Reviewing & Verifying Beneficiary Profiles	S	Working
Updating Beneficiary Status	S	Working
Managing Beneficiary Profiles & Records	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix Table 14. Module: Reports and Dashboard

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Admin Side		
Viewing Real-Time Impact Dashboard	S	Working
Generating PDF Operational Reports	S	Working
Generating Digital Participation Certificates	S	Working
Exporting Project Analytics	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix Table 15. Module: E-Commerce

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Regular User Side		
Browsing Community Products Catalog	S	Working
Placing an Order	S	Working
Uploading Proof of Payment (GCash/Bank)	S	Working
Viewing Transaction & Order History	S	Working
Admin Side		
Adding / Updating Product Items & Stock	S	Working
Updating Customer Order Status	S	Working
Verifying Payment Receipts	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix Table 16. Module: Community Engagement

FUNCTION	SATISFACTORY (S) /UNSATISFACTORY (U)	NOTE
Regular User Side		
Viewing Announcements & Event Calendar	S	Working
Submitting Community Feedback	S	Working
Browsing Community Forum & Discussion Boards	S	Working
Admin Side		
Posting New Announcements	S	Working
Reviewing & Managing Submitted Feedback	S	Working



Tested by: CHRISHAWNE OMER E. BERTILLO

Appendix 12. Curriculum Vitae



CHRISHAWNE OMER E. BERTILLO

SUMMARY

Passionate about building functional, user-friendly web apps, with a foundation in frontend development and database management. Hardworking and eager to improve technical skills.

CONTACT

(+63) 994 433 5348

chrisshawnebertillo@gmail.com

Block 4 Purok 1, Bayanan Muntinlupa City

EDUCATION

Bachelor of Science in Information Technology

Cavite State University - Bacoor Campus

2022 - Present

TVL - Information and Communications Technology

Lyceum of Alabang

2020 - 2022

SKILLS

Technical Skills

- HTML, CSS and Javascript
- Python - Django
- Basic Php and MySQL
- Basic Networking

Soft Skills

- Hardworking
- Always willing to learn

ACADEMIC PROJECTS

2022 – 2023

Responsive Personal Portfolio

- Developed responsive web interfaces using HTML, CSS, and JavaScript
- Created a personal portfolio optimized for desktop and mobile devices

2023 – 2024

Point of Sale (POS) System

- Built a POS system for product management and transaction recording

2023 – 2024

CRM System

- Developed a rental system allowing users to book and track motorbikes using Phyton and Django

2025 – Present

Community Management System

- Currently developing a web-based system for volunteer, event, and donation management

CERTIFICATIONS

- SoloLearn – Web Development Courses
- Cisco Networking Academy - Digital Awareness (2025)

Appendix 12. Continued

Gerod S. Aruta



09078790960 | gerodarutagh@gmail.com |
Blk 28 Lot 5 Valentine's St. Holiday Homes, Biclatan, General Trias, Cavite

SUMMARY

Motivated Information Technology student with hands-on experience building web-based systems through academic projects. Seeking opportunities to apply technical knowledge in real-world settings, collaborate with industry professionals, and further develop practical skills through experiential learning.

ACADEMIC PROJECTS

2024 **Enrollment/Registration System**

Developed an enrollment and registration system that automates student record management, subject selection, and registration processes. Improved data organization and reduced manual processing time.

2023 **Point Of Sale System**

Designed and implemented a food point-of-sale system that streamlines order management, automates sales computation, and facilitates payment processing. Enhanced transaction efficiency and minimized errors in daily operations.

EDUCATION

Cavite State University Bacoor Campus
2022 - Present

Bachelor of Science in Information Technology
City of Bacoor, Cavite 4102, Philippines

Senior Highschool in Progressive
2020 - 2022

TVL - Information and Communications Technology
Progressive 3, Anthrium St, Bacoor, Cavite

SOFT & TECHICAL SKILLS

Leadership & Team Management
Creativity & Adaptability
Problem-Solving
Computer Setup & Peripheral Installation

Frontend Development:
HTML5, CSS3, JavaScript
Backend Development:
Python, Django
Database Management: MySQL
PHP (Basic Knowledge)

ADDITIONAL INFORMATION

Certifications: Computer System Servicing NC II under Joint Delivery Voucher Program (2022),
Cisco Networking Academy - Digital Awareness (2025)

Appendix 12. Continued



JAKE R. VILLAGRACIA

1471 CAMANSI ST. CAA LPC
jakeraro44@gmail.com
09519018408

SUMMARY

Dedicated BSIT student and aspiring IT professional with hands-on experience in hardware troubleshooting, and data management. Eager to bring strong problem-solving skills and technical adaptability to a dynamic organization to help drive efficient IT solutions.

ACADEMIC PROJECT

BASIC EDITING IN MULTIMEDIA 2023

POINT OF SALE SYSTEM 2022
We developed a simple food POS system record an order,tracking and managing stock

TROUBLESHOOTING 2022
Basic troubleshooting in Computer

INTERN EXPERIENCES

Gram Industry, Inc. | Las Piñas City | March 3, 2026 – June 20, 2026
IT Operations & Technical Specialist

- **Spearheaded hardware refurbishment, diagnostic troubleshooting, and OS formatting to optimize employee desktop workstations and reduce downtime.**
- **Managed e-commerce catalog operations and digital inventory records using Microsoft Excel and customized data structures.**
- **Designed precise AutoCAD 2D drafting schematics and equipment layouts for industrial machinery projects.**
- **Resolved staff IT tickets, software bugs, and system configuration issues across office hardware**

TECHNICAL SKILS

- Languages: HTML, CSS, Python
- Troubleshooting
- Editing

SKILLS

- Teamwork
- Adaptability And Willing to Learn
- Time management
- Good Communication

EDUCATION

CAVITE STATE UNIVERSITY BACOOR - CAMPUS 2022-Present
Bachelor of Science in Information Technology

SAINT FRANCIS OF ASSISI COLLAGE 2020- 2022
TVL-ICT

CAA NATIONAL HIGH SCHOOL 2016 - 2010
CAA ELEMENTARY SCHOOL - MAIN 2010 - 2016

PERSONAL INFORMATION

AGE: 22
DATE OF BIRTH: AUGUST 03, 2003
STATUS: SINGLE
NATIONALITY: FILIPINO
RELIGION: CATHOLIC
LANGUAGES: ENGLISH, FILIPINO,

Appendix 12. Continued

Mario F. Bullecer

Address: Queen's Row, Central, Bacoor, Cavite
Email: mariobullecerrs@gmail.com
Phone: (+63) 09463828632



OBJECTIVE

To obtain an on-the-job training (OJT) opportunity in a company where I can apply my knowledge as a BSIT student, enhance my technical and professional skills, and gain practical experience in preparation for my future career.

Technical Skills

- **Programming:** Python
- **Web Development:** HTML, CSS
- **Tools & Platforms:** GitHub, Visual Studio Code
- **Operating Systems:** Windows OS
- **Hardware & Troubleshooting:** Computer setup, peripheral installation, basic hardware troubleshooting
- **Productivity Tools:** Microsoft Office (Word, PowerPoint)

Soft Skills

- Communication and Teamwork
- Multitasking and Time Management
- Adaptable and Willing to Learn

EDUCATION

Elementary School 2005 - 2011
Queen's Row Elementary School

High School 2011 – 2017
Eastern Bacoor National

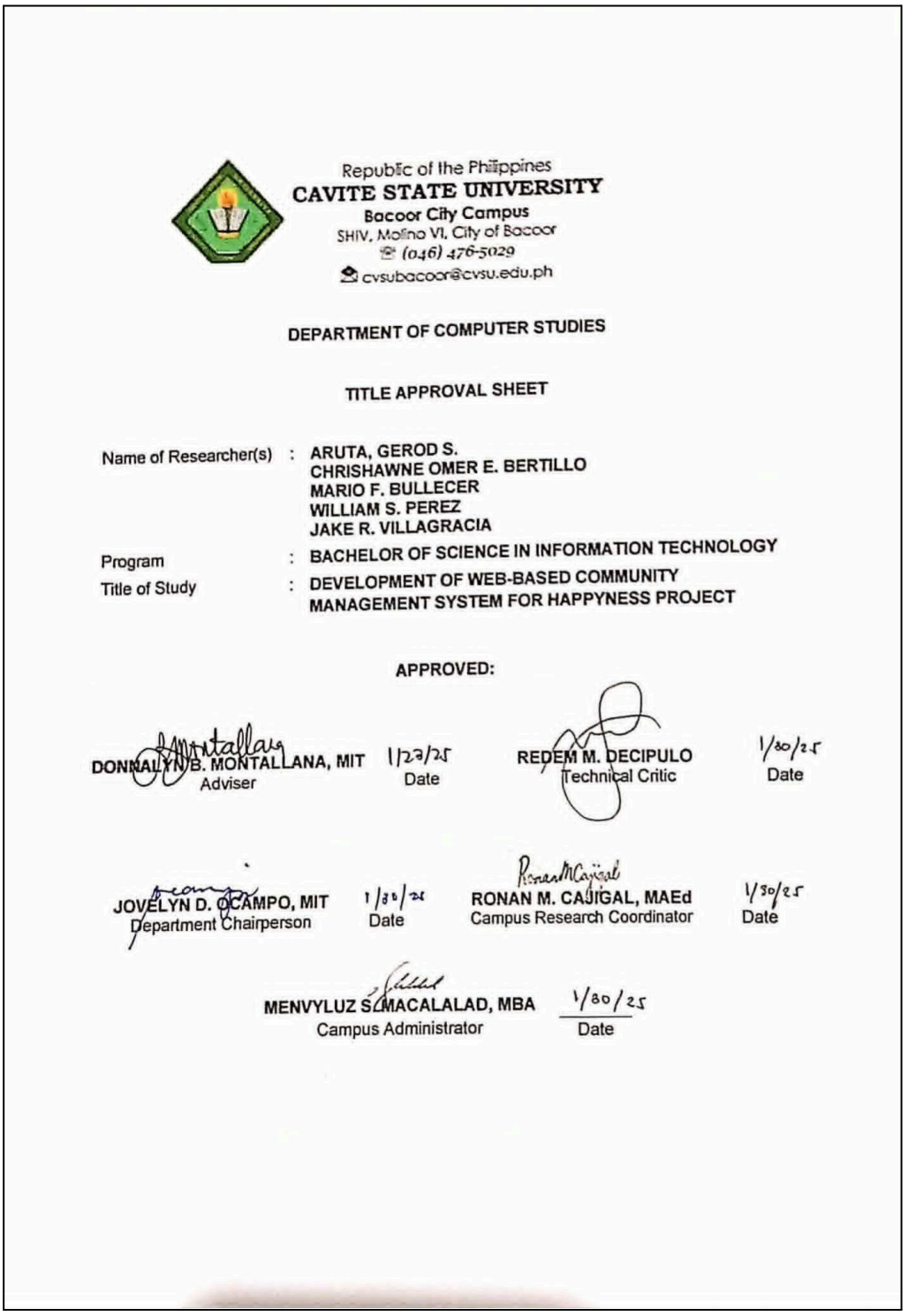
Senior High School 2018 – 2020
San Nicolas Senior High School

Cavite State University – Bacoor City Campus 2022 - present

CERTIFICATIONS

- **National Certificate II** (NC II) – Computer Systems Servicing
- **Digital Awareness** – Cisco Networking Academy
  Issued by Networking Academy (Cisco)

Appendix 13. Title Approval Sheet



Republic of the Philippines
CAVITE STATE UNIVERSITY
Bacoor City Campus
SHIV, Molino VI, City of Bacoor
(046) 476-5029
cvsubacoor@cvsu.edu.ph

DEPARTMENT OF COMPUTER STUDIES

TITLE APPROVAL SHEET

Name of Researcher(s) : **ARUTA, GEROD S.**
CHRISHAWNE OMER E. BERTILLO
MARIO F. BULLECER
WILLIAM S. PEREZ
JAKE R. VILLAGRACIA

Program : **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**

Title of Study : **DEVELOPMENT OF WEB-BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT**

APPROVED:

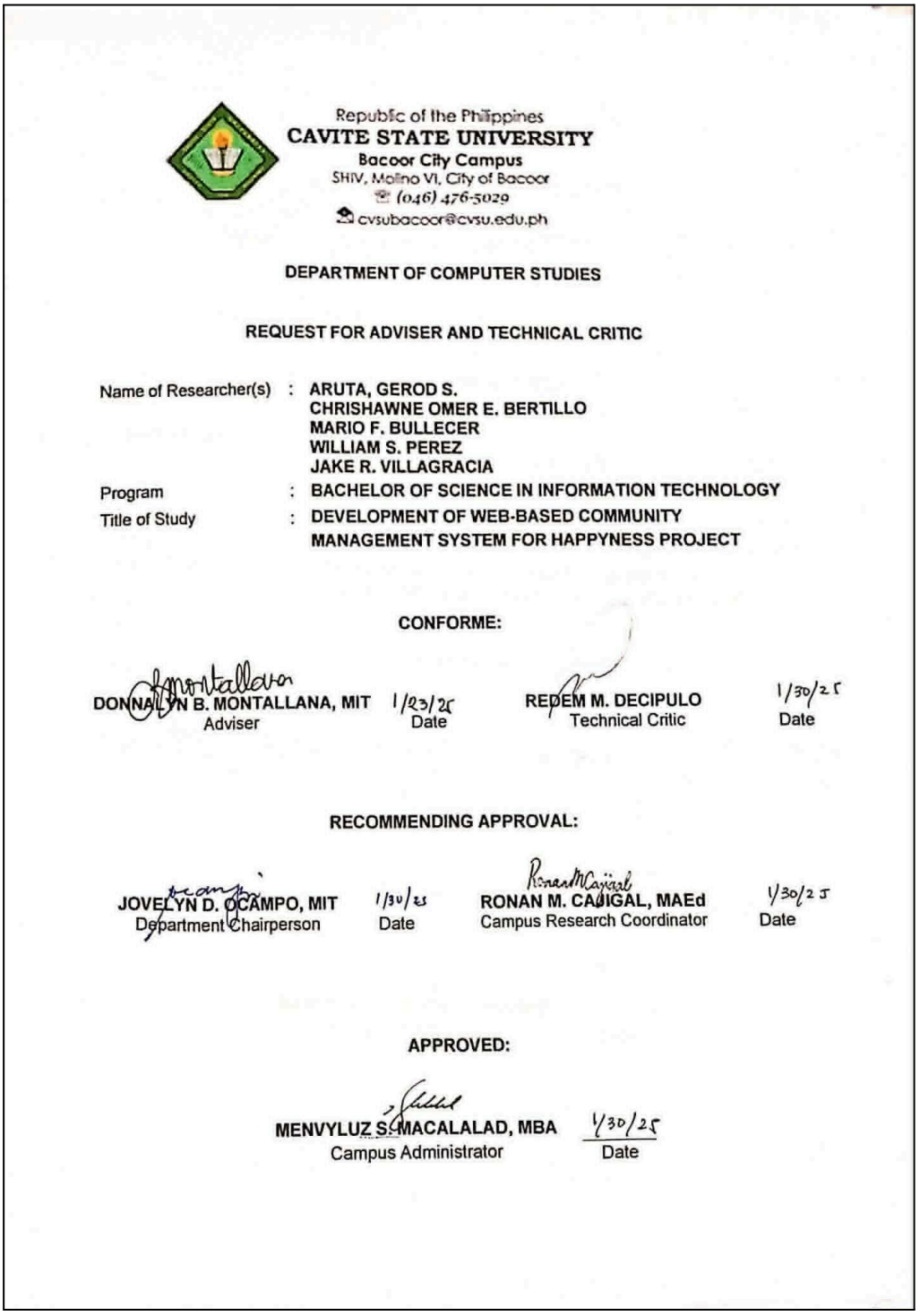
DONNALYN B. MONTALLANA, MIT — Adviser — 1/23/25 Date
REDEM M. DECIPULO — Technical Critic — 1/30/25 Date

JOVELYN D. OCAMPO, MIT — Department Chairperson — 1/30/25 Date
RONAN M. CAJIGAL, MAEd — Campus Research Coordinator — 1/30/25 Date

MENVYLUZ S. MACALALAD, MBA — Campus Administrator — 1/30/25 Date

Appendix Figure 8. Title Approval Sheet

Appendix 14. Request for Thesis Adviser and Technical Critic



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DEPARTMENT OF COMPUTER STUDIES

REQUEST FOR ADVISER AND TECHNICAL CRITIC

Name of Researcher(s) : ARUTA, GEROD S.
CHRISHAWNE OMER E. BERTILLO
MARIO F. BULLECER
WILLIAM S. PEREZ
JAKE R. VILLAGRACIA

Program : BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

Title of Study : DEVELOPMENT OF WEB-BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT

CONFORME:

DONNALYN B. MONTALLANA, MIT — Adviser — 1/23/25 Date
REDEM M. DECIPULO — Technical Critic — 1/30/25 Date

RECOMMENDING APPROVAL:

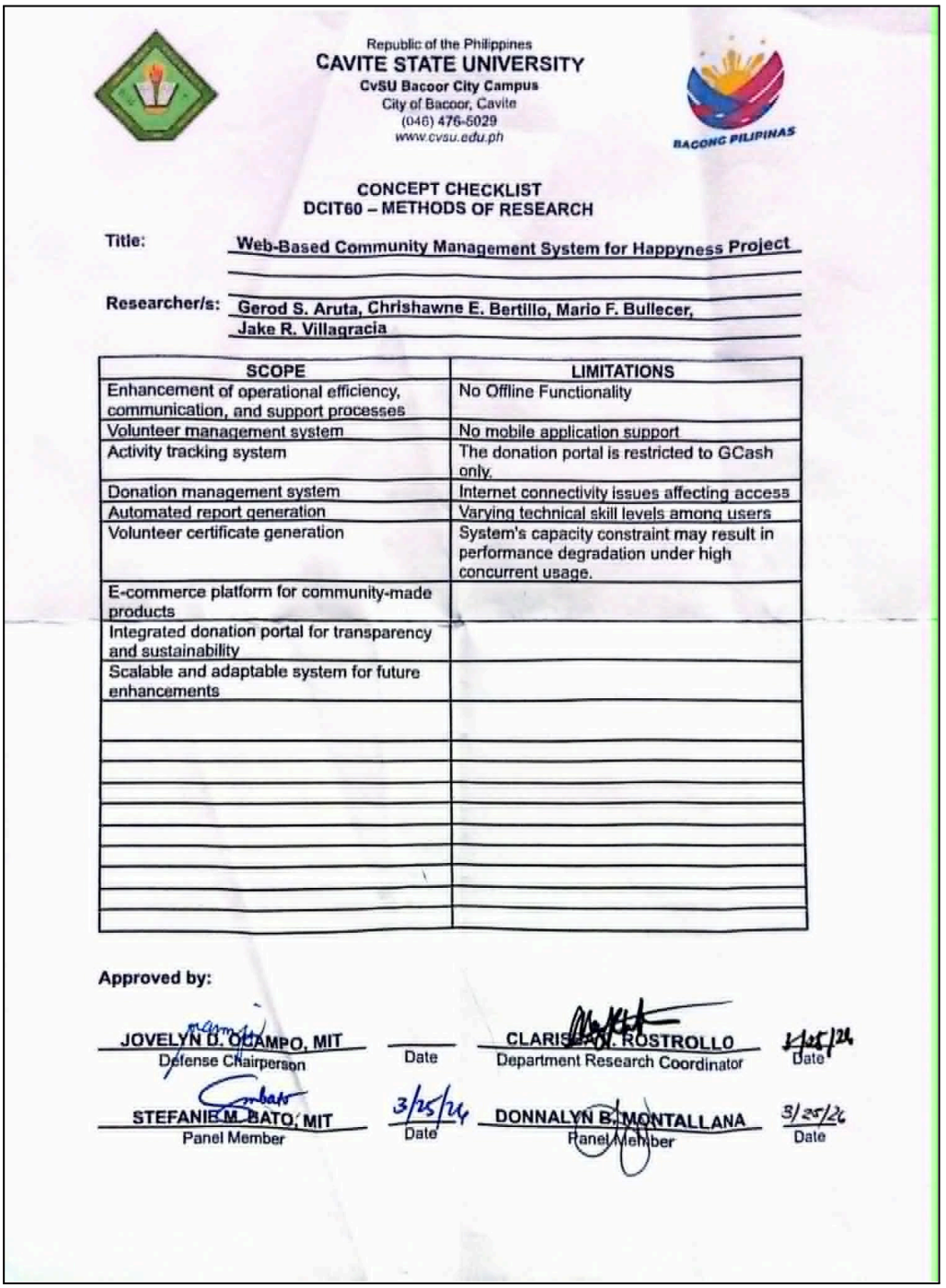
JOVELYN D. OCAMPO, MIT — Department Chairperson — 1/30/25 Date
RONAN M. CAJIGAL, MAEd — Campus Research Coordinator — 1/30/25 Date

APPROVED:

MENVYLUZ S. MACALALAD, MBA — Campus Administrator — 1/30/25 Date

Appendix Figure 9. Request for Thesis Adviser and Technical Critic

Appendix 15. Concept Checklist



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BAGONG PILIPINAS

CONCEPT CHECKLIST
DCIT60 – METHODS OF RESEARCH

Title: Web-Based Community Management System for Happyness Project

Researcher/s: Gerod S. Aruta, Chrishawne E. Bertillo, Mario F. Bullecer, Jake R. Villagracia

SCOPE	LIMITATIONS
Enhancement of operational efficiency, communication, and support processes	No Offline Functionality
Volunteer management system	No mobile application support
Activity tracking system	The donation portal is restricted to GCash only.
Donation management system	Internet connectivity issues affecting access
Automated report generation	Varying technical skill levels among users
Volunteer certificate generation	System's capacity constraint may result in performance degradation under high concurrent usage.
E-commerce platform for community-made products	
Integrated donation portal for transparency and sustainability	
Scalable and adaptable system for future enhancements	

Approved by:

JOVELYN D. OCAMPO, MIT
Defense Chairperson
Date

CLARISSA N. ROSTROLLO
Department Research Coordinator
3/25/26
Date

STEFANIE M. BATO, MIT
Panel Member
3/25/26
Date

DONNALYN B. MONTALLANA
Panel Member
3/25/26
Date

Appendix Figure 10. Concept Checklist

Appendix 16. Certificate of Statistical Analysis

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BAGONG PILIPINAS

October 29, 2026

CERTIFICATION

This is to certify that the thesis entitled: **"THE DEVELOPMENT OF WEB-BASED COMMUNITY MANAGEMENT SYSTEM FOR HAPPYNESS PROJECT"** of **Gerod S. Aruta, Chrishawne Omer E. Bertillo, Mario F. Bullecer, and Jake R. Villagracia** has been carefully examined to the best of my ability to satisfy its purpose and needed information by the undersigned as Statistician.

ZANNIE J. GAMUYAO, MSM
Associate Professor V
Statistician